

Chapter 3: Transmission Lines

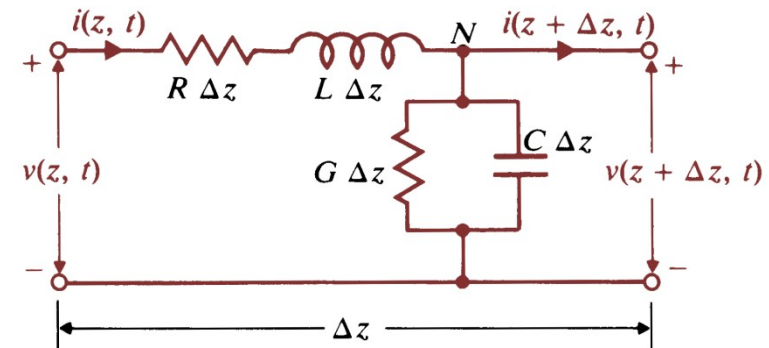
Part 3: General Transmission Line Equations

Electric Circuits vs. TL

- Shall now derive equations that govern general 2-conductor uniform transmission lines, i.e. parallel-plate, 2-wire, & coaxial TLs
- Differences of TL from electrical circuits (EC):
 - physical dimensions of ECs much smaller than wavelength, whereas TLs usually over considerable fraction of λ , or even many multiples of it
 - EC elements considered **discrete**, described by **lumped parameters**, no spatial variation over them, whereas TLs ***distributed-parameter network***, parameters distributed throughout length

4 Characterizing Parameters

- Consider differential length Δz of a TL described by following 4 parameters:
 - ✓ R , resistance per unit length (both conductors), in Ω/m
 - ✓ L , inductance per unit length (both conductors), in H/m
 - ✓ G , conductance per unit length (S/m)
 - ✓ C , capacitance per unit length (F/m)
- At z & $z + \Delta z$: instantaneous
 - ✓ voltages: $v(z, t)$ & $v(z + \Delta z, t)$
 - ✓ currents: $i(z, t)$ & $i(z + \Delta z, t)$
- R & L : series elements, G & C : shunt elements



Kirchoff's Voltage Law

$$v(z, t) - R\Delta z \cdot i(z, t) - L\Delta z \frac{\partial i(z, t)}{\partial t} - v(z + \Delta z, t) = 0 \quad (3.1a)$$

$$\Rightarrow \underbrace{-\frac{v(z + \Delta z, t) - v(z, t)}{\Delta z}}_{\substack{\text{as } \Delta z \rightarrow 0 \\ \frac{\partial v(z, t)}{\partial z}}} = Ri(z, t) + L \frac{\partial i(z, t)}{\partial t} \quad (3.1b)$$

$$\therefore \boxed{\frac{\partial v(z, t)}{\partial z} = Ri(z, t) + L \frac{\partial i(z, t)}{\partial t}} \quad (3.2)$$

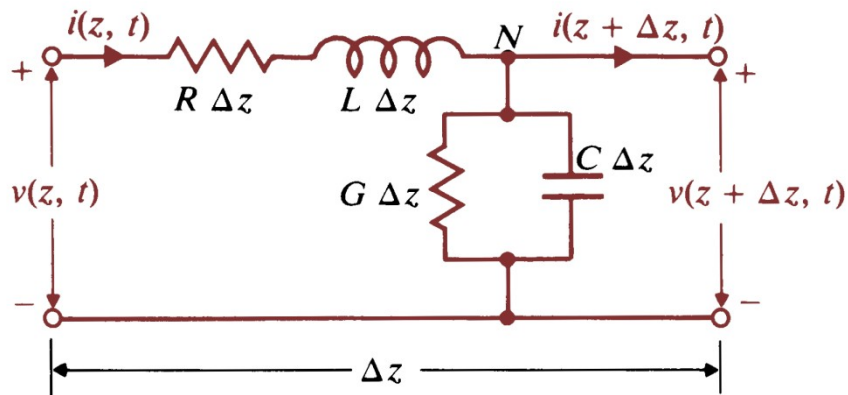


FIGURE 9-5
Equivalent circuit of a differential length Δz of a two-conductor transmission line.

Kirchoff's Current Law to node N .

$$i(z, t) - G\Delta z \cdot v(z + \Delta z, t) - C\Delta z \frac{\partial v(z + \Delta z, t)}{\partial t} - i(z + \Delta z, t) = 0 \quad (3.3a)$$

$$\Rightarrow \underbrace{-\frac{i(z + \Delta z, t) - i(z, t)}{\Delta z}}_{\substack{\text{as } \Delta z \rightarrow 0 \\ \frac{\partial i(z, t)}{\partial z}}} = G \underbrace{v(z + \Delta z, t)}_{\substack{\text{as } \Delta z \rightarrow 0 \\ v(z, t)}} + C \underbrace{\frac{\partial v(z + \Delta z, t)}{\partial t}}_{\substack{\text{as } \Delta z \rightarrow 0 \\ \frac{\partial v(z, t)}{\partial t}}} \quad (3.3b)$$

$$\therefore \boxed{\frac{\partial i(z, t)}{\partial z} = Gv(z, t) + C \frac{\partial v(z, t)}{\partial t}} \quad (3.4)$$

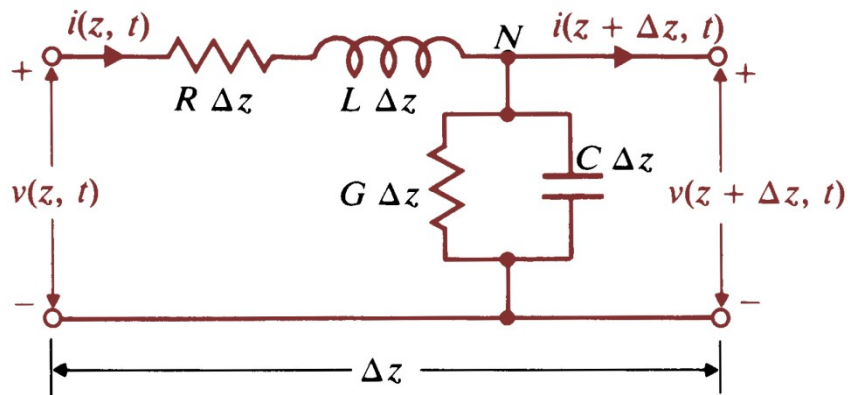


FIGURE 9-5

Equivalent circuit of a differential length Δz of a two-conductor transmission line.

Telegrapher's Equations

- Eqs. (3.2) & (3.4): pair of 1st order partial differential equations (d.e.) in $v(z, t)$ & $i(z, t) \rightarrow$ *general transmission-line equations* (sometimes also called: *telegrapher's equations*)
- For harmonic time-dependence, use of phasors simplifies TL-equations to ordinary d.e.:

$$v(z, t) = \operatorname{Re} \left[V(z) e^{j\omega t} \right] \quad (3.5a)$$

$$i(z, t) = \operatorname{Re} \left[I(z) e^{j\omega t} \right] \quad (3.5b)$$

Time-Harmonic TL Equations

- Substitute these (3.5a) & (3.5b) into (3.2) & (3.4)

$$-\frac{dV(z)}{dz} = (R + j\omega L)I(z) \quad (3.6a)$$

$$-\frac{dI(z)}{dz} = (G + j\omega C)V(z) \quad (3.6b)$$

- These (3.6a) & (3.6b): *time-harmonic transmission-line equations* $\rightarrow$ reduce to (2.11) & (2.14) under lossless conditions ($R = G = 0$)

Part 3.1: Wave Characteristics on an Infinite Transmission Line

Solve $V(z)$ & $I(z)$; atten & phase const

- Coupled time-harmonic TL equations (3.6a) & (3.6b) combined to solve for $V(z)$ & $I(z)$:

$$\text{Diff (3.6a) w.r.t. } z : \frac{d^2 V(z)}{dz^2} = -(R + j\omega L) \underbrace{\frac{dI(z)}{dz}}_{\substack{\text{from (3.6b)} \\ -(G + j\omega C)V(z)}} \stackrel{\text{define}}{=} \gamma^2 V(z) \quad (3.7a)$$

$$\text{Diff (3.6b) w.r.t. } z : \frac{d^2 I(z)}{dz^2} = -(G + j\omega C) \underbrace{\frac{dV(z)}{dz}}_{\substack{\text{from (3.6a)} \\ -(R + j\omega L)I(z)}} \stackrel{\text{define}}{=} \gamma^2 I(z) \quad (3.7b)$$

- Propagation constant γ :
 - ✓ real part α : attenuation constant (Np/m)
 - ✓ imaginary part β : phase constant (rad/m)

$$\gamma = \underbrace{\alpha}_{\substack{\text{atten} \\ \text{const} \\ (\text{Np/m})}} + j \underbrace{\beta}_{\substack{\text{phase} \\ \text{const} \\ (\text{rad/m})}} = \sqrt{(R + j\omega L)(G + j\omega C)} \quad (\text{m}^{-1}) \quad (3.8)$$

Traveling-Wave Solutions

- Solutions of (3.7a) & (3.7b):
 - ✓ $V(z) = V^+(z) + V^-(z) = V_0^+ e^{-\gamma z} + V_0^- e^{+\gamma z},$ (3.9a)
 - ✓ $I(z) = I^+(z) + I^-(z) = I_0^+ e^{-\gamma z} + I_0^- e^{+\gamma z},$ (3.9b)
- + & – superscripts: +z & –z directions of wave travel
- Wave amplitudes (V_0^+, I_0^+) & (V_0^-, I_0^-) related by (3.6a) & (3.6b) – next slides working ...

Working (1)

$$\frac{d}{dz} \overbrace{V(z)}^{V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}} = -\gamma (V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z}) \quad (3.10a)$$

$$\frac{d}{dz} \overbrace{I(z)}^{I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}} = -\gamma (I_0^+ e^{-\gamma z} - I_0^- e^{\gamma z}) \quad (3.10b)$$

$$\begin{aligned} & \text{sub} \quad \frac{-\frac{dV(z)}{dz}}{(3.10a)} \quad \overbrace{\gamma (V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z})}^{I(z)} = (R + j\omega L) \overbrace{(I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z})}^{I(z)} \Rightarrow V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z} = \frac{(R + j\omega L)(I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z})}{\gamma} \\ & \text{into} \quad (3.6a) \end{aligned} \quad (3.11a)$$

$$\begin{aligned} & \text{sub} \quad \frac{\frac{dI(z)}{dz}}{(3.10b)} \quad \overbrace{\gamma (I_0^+ e^{-\gamma z} - I_0^- e^{\gamma z})}^{V(z)} = (G + j\omega C) \overbrace{(V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z})}^{V(z)} \Rightarrow V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z} = \frac{\gamma (I_0^+ e^{-\gamma z} - I_0^- e^{\gamma z})}{G + j\omega C} \\ & \text{into} \quad (3.6b) \end{aligned} \quad (3.11b)$$

$$(3.11a) + (3.11b) : 2V_0^+ e^{-\gamma z} = \frac{(R + j\omega L)(I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z})}{\gamma} + \frac{\gamma (I_0^+ e^{-\gamma z} - I_0^- e^{\gamma z})}{G + j\omega C}$$

$$\begin{aligned} & \overbrace{\frac{(R + j\omega L)(I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z})}{\gamma} + \frac{\gamma (I_0^+ e^{-\gamma z} - I_0^- e^{\gamma z})}{G + j\omega C}}^{2\gamma I_0^+ e^{-\gamma z}} \\ & = \frac{\overbrace{(R + j\omega L)(G + j\omega C)(I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z})}^{\gamma^2} + \gamma^2 (I_0^+ e^{-\gamma z} - I_0^- e^{\gamma z})}{\gamma (G + j\omega C)} = \frac{2\gamma I_0^+ e^{-\gamma z}}{G + j\omega C} \end{aligned}$$

$$\Rightarrow V_0^+ = \frac{\gamma I_0^+}{G + j\omega C} \Rightarrow \frac{V_0^+}{I_0^+} = \frac{\gamma}{G + j\omega C} \quad (3.12a)$$

Working (2)

$$\begin{aligned}
 (3.11b) - (3.11a) : 2V_0^- e^{\gamma z} &= \frac{\gamma(I_0^+ e^{-\gamma z} - I_0^- e^{\gamma z})}{G + j\omega C} - \frac{(R + j\omega L)(I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z})}{\gamma} \\
 &= \frac{\overbrace{\gamma^2(I_0^+ e^{-\gamma z} - I_0^- e^{\gamma z}) - \overbrace{(R + j\omega L)(G + j\omega C)}^{\gamma^2}(I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z})}^{-2\gamma I_0^- e^{\gamma z}}}{\cancel{\chi}(G + j\omega C)} = -\frac{2\gamma I_0^- e^{\gamma z}}{G + j\omega C} \\
 \Rightarrow V_0^- &= -\frac{\gamma I_0^-}{G + j\omega C} \Rightarrow \frac{V_0^-}{I_0^-} = -\frac{\gamma}{G + j\omega C} \quad (3.12b)
 \end{aligned}$$

Working (3)

From (3.11a):

$$V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z} = \frac{(R + j\omega L) I_0^+}{\gamma} e^{-\gamma z} + \frac{(R + j\omega L) I_0^-}{\gamma} e^{\gamma z}$$

$$\begin{aligned} \text{equate} \\ \text{coeff} \\ \text{of} \\ e^{-\gamma z} \end{aligned} \Rightarrow V_0^+ = \frac{(R + j\omega L) I_0^+}{\gamma} \Rightarrow \frac{V_0^+}{I_0^+} = \underbrace{\frac{R + j\omega L}{\gamma}}_{\substack{\text{consistent} \updownarrow \text{with (3.8)} \\ \gamma^2 = (R + j\omega L)(G + j\omega C)}} \stackrel{\text{from (3.12a)}}{=} \frac{\gamma}{G + j\omega C} \quad (3.13a)$$

$$\begin{aligned} \text{equate} \\ \text{coeff} \\ \text{of} \\ e^{\gamma z} \end{aligned} \Rightarrow V_0^- = -\frac{(R + j\omega L) I_0^-}{\gamma} \Rightarrow \frac{V_0^-}{I_0^-} = -\underbrace{\frac{R + j\omega L}{\gamma}}_{\substack{\text{consistent} \updownarrow \text{with (3.8)} \\ \gamma^2 = (R + j\omega L)(G + j\omega C)}} \stackrel{\text{from (3.12b)}}{=} -\frac{\gamma}{G + j\omega C} \quad (3.13b)$$

Combining (3.12a) & (3.12b):

$$\overbrace{\frac{V_0^+}{I_0^+} = -\frac{V_0^-}{I_0^-}}^{\text{combine (3.12a) \& (3.12b)}} \stackrel{\text{use (3.13a)}}{=} \frac{\gamma}{G + j\omega C} \stackrel{\text{\&(3.13b) above}}{=} \frac{R + j\omega L}{\gamma} \quad (3.14)$$

Characteristic Impedance

- Ratio of voltage & current at any z for infinitely-long line independent of $z \rightarrow$ called: characteristic impedance Z_0 of line

$$Z_0 = \frac{V_0^+}{I_0^+} = -\frac{V_0^-}{I_0^-} \stackrel{\substack{\text{combine} \\ (3.12a) \ \& \ (3.12b)}}{=} \stackrel{\substack{\text{use} \\ (3.13a) \ \& \ (3.13b) \\ \text{above}}}{=} \frac{\gamma}{G + j\omega C} = \frac{\sqrt{(R+j\omega L)(G+j\omega C)}}{G + j\omega C} = \frac{\sqrt{\frac{R+j\omega L}{G+j\omega C}}}{\underbrace{\gamma}_{\substack{\text{from } \uparrow (3.8)}}} = \sqrt{\frac{R+j\omega L}{G+j\omega C}} \quad (\Omega) \quad (3.15)$$

- Note: γ & Z_0 are characteristic properties of a TL whether or not infinitely-long

Example: Analogy between TL & Plane Waves in Lossy Medium (1)

Demonstrate analogy between wave characteristics on TL & uniform plane waves in lossy medium.

Solution

In lossy medium with complex parameters: $\varepsilon_c = \varepsilon' - j\varepsilon''$ & $\mu_c = \mu' - j\mu''$, Maxwell's curl equations: (7.14) & (7.15) Part 7 of Chapter 1 become:

$$\nabla \times \vec{E} = -j\omega(\mu' - j\mu'')\vec{H} \quad (3.16a)$$

$$\nabla \times \vec{H} = j\omega(\varepsilon' - j\varepsilon'')\vec{E} \quad (3.16b)$$

Assume uniform plane wave characterized by E_x that varies only with z : above (3.16a) reduces to [see (1.9) of Chapter 2, repeated: $H_y^+ = -\frac{1}{j\omega\mu_0} \frac{\partial E_x^+}{\partial z}$]:

$$-\frac{\partial E_x(z)}{\partial z} = j\omega(\mu' - j\mu'')H_y(z) = \omega(\mu'' + j\mu')H_y(z) \quad (3.17a)$$

Example: Analogy between TL & Plane Waves in Lossy Medium (2)

$$\nabla \times \vec{H} = \begin{pmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{pmatrix} \times \begin{pmatrix} H_x \\ H_y \\ H_z \end{pmatrix} = \begin{pmatrix} \cancel{\partial H_z / \partial y} - \partial H_y / \partial z \\ \partial \cancel{H_x} / \partial z - \cancel{\partial H_z} / \partial x \\ \partial H_y / \cancel{\partial x} - \cancel{\partial H_x} / \partial y \end{pmatrix} \stackrel[\text{(3.16b) above}]{\text{from}} = j\omega(\epsilon' - j\epsilon'') \begin{pmatrix} E_x^+ \\ E_y^+ \\ E_z^+ \end{pmatrix}$$

$$\therefore E_x = -\frac{1}{j\omega(\epsilon' - j\epsilon'')} \frac{\partial H_y}{\partial z}; \quad \& \quad E_y \equiv E_z \equiv 0$$

$$\Rightarrow -\frac{\partial H_y(z)}{\partial z} = \omega(\epsilon'' + j\epsilon') E_x \quad \textbf{(3.17b)}$$

in which all slashes (forward & backward) denote cancellation to zero;

- forward slashes (/) due to vanishing partial derivatives with respect to x & y ($\because$ independence of them),
- whereas backward slashes (\) $\because$ only $H_y \neq 0$.

Example: Analogy between TL & Plane Waves in Lossy Medium (3)

Compare these (3.17a) & (3.17b) with (3.6a) & (3.6b), recognize analogy between uniform plane waves & TL:

$$\begin{aligned}
 -\frac{\partial E_x(z)}{\partial z} &= \omega(\mu'' + j\mu') H_y(z) & -\frac{dV(z)}{dz} &= (R + j\omega L) I(z) & \text{(3.6a)repeated} \\
 -\frac{\partial H_y(z)}{\partial z} &= \omega(\varepsilon'' + j\varepsilon') E_x(z) & -\frac{dI(z)}{dz} &= (G + j\omega C) V(z) & \text{(3.6b)repeated}
 \end{aligned}$$

Combine (3.17a) & (3.17b):

$$\text{Diff (3.17a) w.r.t. } z: \frac{d^2 E_x(z)}{dz^2} = -\omega(\mu'' + j\mu') \underbrace{\frac{dH_y(z)}{dz}}_{\substack{\text{from (3.17b)} \\ -\omega(\varepsilon'' + j\varepsilon') E_x(z)}} \stackrel{\text{define}}{=} \gamma^2 E_x(z) \quad \text{(3.18a)}$$

$$\text{Diff (3.17b) w.r.t. } z: \frac{d^2 H_y(z)}{dz^2} = -\omega(\varepsilon'' + j\varepsilon') \underbrace{\frac{dE_x(z)}{dz}}_{\substack{\text{from (3.17a)} \\ -\omega(\mu'' + j\mu') H_y(z)}} \stackrel{\text{define}}{=} \gamma^2 H_y(z) \quad \text{(3.18b)}$$

being analogous to (3.7a) & (3.7b) earlier

Example: Analogy between TL & Plane Waves in Lossy Medium (4)

- Propagation constant γ of uniform plane wave

$$\gamma = \underbrace{\alpha}_{\substack{\text{atten} \\ \text{const} \\ (\text{Np/m})}} + j \underbrace{\beta}_{\substack{\text{phase} \\ \text{const} \\ (\text{rad/m})}} = \underbrace{\sqrt{\omega^2 (\mu'' + j\mu')(\varepsilon'' + j\varepsilon')}}_{\omega \sqrt{(\mu'' + j\mu')(\varepsilon'' + j\varepsilon')}} \quad (3.19)$$

- Compare this with (3.8) for TL earlier, repeated:

$$\gamma = \underbrace{\alpha}_{\substack{\text{atten} \\ \text{const} \\ (\text{Np/m})}} + j \underbrace{\beta}_{\substack{\text{phase} \\ \text{const} \\ (\text{rad/m})}} = \sqrt{(R + j\omega L)(G + j\omega C)} \quad (\text{m}^{-1}) \quad (3.8)\text{repeated}$$

- Intrinsic impedance of lossy medium (wave impedance of plane wave) [see (1.23) Chapter 2]

$$\eta_c = \sqrt{\frac{\mu'' + j\mu'}{\varepsilon'' + j\varepsilon'}} = \sqrt{\frac{j(\mu' - j\mu'')}{j(\varepsilon' - j\varepsilon'')}} = \sqrt{\frac{\mu_c}{\varepsilon_c}} \quad (3.20)$$

analogous to (3.15)

Three Special Cases

Case 1: Lossless ($R = 0, G = 0$)

$$\gamma = \sqrt{(\underbrace{R + j\omega L}_{=0})(\underbrace{G + j\omega C}_{=0})} = \sqrt{-\omega^2 LC} = j\omega\sqrt{LC}$$

- Propagation constant [from (3.8) with $R = G = 0$]: $\gamma = \alpha + j\beta = j\omega\sqrt{LC}$ (3.21)
 - ✓ $\alpha = 0$ (3.21a)
 - ✓ $\beta = \omega\sqrt{LC} \rightarrow$ linear function of ω . (3.21b)
- Phase velocity [from (2.21)]: $u_p = \omega/\beta = 1/\sqrt{LC}$
 $\rightarrow$ constant (3.22)
- Characteristic impedance [from (3.15)]

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{L}{C}} \stackrel{\text{write}}{=} \underbrace{\sqrt{\frac{L}{C}}}_{R_0} + j \underbrace{0}_{X_0} \quad (3.23)$$

Three Special Cases

Case 2: Low Loss ($R \ll \omega L, G \ll \omega C$) #1

Propagation constant [from (3.8)] $\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$

$$\begin{aligned} \gamma = \alpha + j\beta &= \sqrt{(R + j\omega L)(G + j\omega C)} = \sqrt{(j\omega L)\left(1 + \frac{R}{j\omega L}\right)(j\omega C)\left(1 + \frac{G}{j\omega C}\right)} \\ &= j\omega\sqrt{LC}\left(1 + \frac{R}{j\omega L}\right)^{\frac{1}{2}}\left(1 + \frac{G}{j\omega C}\right)^{\frac{1}{2}} \quad \begin{array}{l} \text{approx by} \\ \text{binomial} \\ \text{expansion} \end{array} \\ &\quad \begin{array}{l} \cong \\ \text{with } \frac{R}{j\omega L} \ll 1 \\ \& \frac{G}{j\omega C} \ll 1 \end{array} j\omega\sqrt{LC}\underbrace{\left(1 + \frac{R}{2j\omega L}\right)\left(1 + \frac{G}{2j\omega C}\right)}_{1 + \frac{G}{2j\omega C} + \frac{R}{2j\omega L} - \underbrace{\frac{RG}{4\omega^2 LC}}_{\substack{\text{negligible} \\ \text{compared to} \\ \text{leading terms}}}} \end{aligned}$$

$$\cong j\omega\sqrt{LC}\left[1 + \frac{1}{2j\omega}\left(\frac{R}{L} + \frac{G}{C}\right)\right] = \underbrace{\frac{j\omega}{2j\omega}\sqrt{LC}\left(\frac{R}{L} + \frac{G}{C}\right)}_{R\sqrt{\frac{C}{L}} + G\sqrt{\frac{L}{C}}} + j\omega\sqrt{LC} \quad (3.24)$$

check with (3.3-7) later

$$(3.24a) \Rightarrow \alpha \cong \frac{1}{2}\left(R\sqrt{\frac{C}{L}} + G\sqrt{\frac{L}{C}}\right) \quad \& \quad (3.24b) \quad \beta \cong \omega\sqrt{LC} \quad \text{approx linear function of } \omega. \quad ^{21}$$

Three Special Cases

Case 2: Low Loss ($R \ll \omega L$, $G \ll \omega C$) #2

Phase velocity [from (2.21)]:

$$u_p = \frac{\omega}{\beta} \overset{\substack{\text{use} \\ (3.24b)}}{\underset{\beta \cong \omega \sqrt{LC}}{\approx}} \frac{1}{\sqrt{LC}} \quad (3.25)$$

Three Special Cases

Case 2: Low Loss ($R \ll \omega L, G \ll \omega C$) #3

Characteristic impedance [from (3.15)]

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{j\omega L \left(\frac{R}{j\omega L} + 1 \right)}{j\omega C \left(\frac{G}{j\omega C} + 1 \right)}} = \sqrt{\frac{L}{C}} \frac{\overbrace{\left(1 + \frac{R}{j\omega L} \right)^{\frac{1}{2}}}^{\substack{\text{approxim by } \uparrow \text{binom expan} \\ 1 + \frac{R}{2j\omega L}}}}{\underbrace{\left(1 + \frac{G}{j\omega C} \right)^{\frac{1}{2}}}_{\substack{\text{approxim by } \uparrow \text{binom expan} \\ 1 + \frac{G}{2j\omega C}}}} \cong \sqrt{\frac{L}{C}} \frac{\overbrace{\frac{2j\omega L + R}{2j\omega L}}^{2j\omega L \left(1 + \frac{R}{2j\omega L} \right)}}{\cancel{2j\omega C}} = \sqrt{\frac{C}{L}} \frac{\overbrace{\cancel{2j\omega L} \left(1 + \frac{R}{2j\omega L} \right)}^{\substack{\text{approx } 1 \\ \ll 1}}}{\cancel{2j\omega C} \underbrace{\left(1 + \frac{G}{2j\omega C} \right)}_{\substack{\text{approx } 1 \\ \ll 1}}}$$

$$\cong \sqrt{\frac{L}{C}} \stackrel{\substack{\text{write} \\ \text{as}}}{=} \overbrace{\sqrt{\frac{L}{C}}}^0 R_0 + j \overbrace{X_0}^0$$

(3.26)

$$\therefore R_0 \cong \sqrt{\frac{L}{C}} \quad (3.26a)$$

$$\& X_0 \cong 0 \quad (3.26b)$$

Three Special Cases

Case 3: Distortionless ($R/L = G/C$) #1

Propagation constant [from (3.8)] $\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$

$$\gamma = \alpha + j\beta \quad \begin{matrix} \text{presently} \\ \text{distortion} \\ \text{-less} \end{matrix} \quad \begin{matrix} = \\ \frac{R}{L} = \frac{G}{C} \end{matrix} \quad \sqrt{(R + j\omega L) \left[\underbrace{\left(\frac{RC}{L} \right)}_G + j\omega C \right]} = \sqrt{(R + j\omega L) \left[\frac{C}{L} (R + j\omega L) \right]}$$

$$= \sqrt{\frac{C}{L}} (R + j\omega L)$$

$$\Rightarrow \alpha = R\sqrt{\frac{C}{L}} \quad \& \quad \beta = \omega\sqrt{LC} \quad \text{pure linear function of } \omega.$$

(3.27a) **(3.27b)**

Three Special Cases

Case 3: Distortionless ($R/L = G/C$) #2

Phase velocity [from (2.21)]:

$$u_p = \frac{\omega}{\beta} \stackrel{\text{use}}{=} \frac{1}{\sqrt{LC}} \quad (3.28)$$

$\beta = \omega\sqrt{LC}$ (3.27b)

Characteristic impedance [from (3.15)]

$$\underbrace{\overbrace{\overbrace{R_0 + jX_0}^{\sqrt{L/C}}}}_{Z_0} = \sqrt{\frac{R + j\omega L}{\underbrace{\left(\frac{RC}{L}\right) + j\omega C}_G}} = \sqrt{\frac{L\left(\frac{R}{L} + j\omega\right)}{C\left(\frac{R}{L} + j\omega\right)}} = \sqrt{\frac{L}{C}}$$

(3.29a) $\therefore R_0 = \sqrt{\frac{L}{C}}$ & (3.29b) $X_0 = 0$

Case 3: Distortionless ($R/L = G/C$) #3

Comments about distortionless lines

- Except for nonvanishing attenuation constant, characteristics of distortionless line same as those of lossless line
 - const phase vel. $u_p = 1/\sqrt{LC}$
 - const real charac imped: $Z_0 = R_0 = \sqrt{L/C}$
- Const phase velocity: linear dependence of phase const β on angular freq ω .
- To avoid distortion, all freq components must travel along TL at same velocity $\rightarrow$ satisfied by lossless line, & by $R/L = G/C$ (distortionless line)

Part 3.2: Transmission Line Parameters

Opening Words

- Electrical properties of TL at given freq completely characterized by its 4 distributed parameters: R , L , G , & C .
- Assume conductivity of conductors in TL so high that effect of series resistance on propagation constant negligible
 - waves on line approx TEM
 - may drop R from (3.8):

$$\begin{aligned}\gamma &= \sqrt{(R + j\omega L)(G + j\omega C)} = \sqrt{j\omega LG - \omega^2 LC} \\ &= \sqrt{-\omega^2 LC \left(1 + \frac{G}{j\omega C}\right)} = j\omega \sqrt{LC \left(1 + \frac{G}{j\omega C}\right)}\end{aligned}\quad (3.2-1)$$

Preliminary Relations

- From (3.3) Chapter 2, repeated: $\gamma = j\omega\sqrt{\mu\varepsilon}\sqrt{1 + \frac{\sigma}{j\omega\varepsilon}}$
(3.2-2)
- But $G/C \stackrel{(3.2-3)}{=} \sigma/\varepsilon$ as learned from EM1
- Compare (3.2-1) & (3.2-2): $\boxed{LC = \mu\varepsilon.}$ (3.2-4)
in conformance with (2.18)
- This (3.2-4) useful:
 - if L known for a TL with given medium, C can be found, & vice-versa
 - With C known, can find G from (3.2-3)

Statement

- Velocity of wave propagation on lossless TL, $u_p = 1/\sqrt{LC}$ equals velocity of propagation, $1/\sqrt{\mu\epsilon}$, of unguided plane wave in unbounded space homogeneously filled with medium same as dielectric of line

C, L, G & R Parameters of Two Types of TL

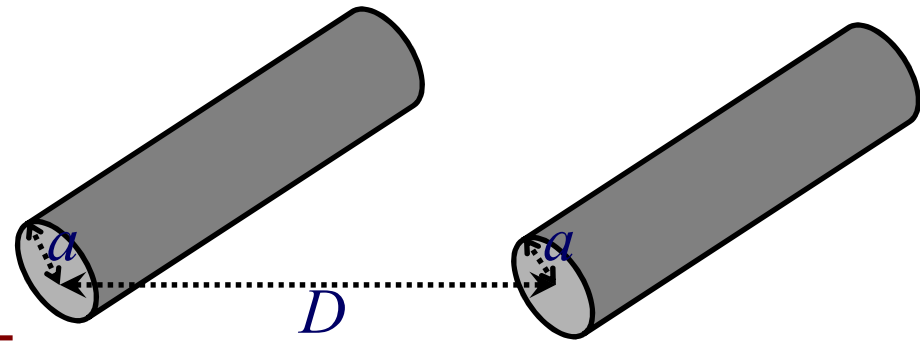
1. Two-wire
2. Coaxial

Formulas shall just be provided here without proof – students are encouraged to refer to textbook for details of derivation

Two-Wire Transmission Line (C , L & G)

Radius of both wires = a .

Separated by distance D .



$$C = \frac{\pi\epsilon}{\cosh^{-1}\left(\frac{D}{2a}\right)}$$

(F/m)

(3.2-5)

$$L = \frac{\mu}{\pi} \cosh^{-1}\left(\frac{D}{2a}\right)$$

(H/m)

(3.2-6)

$$G = \frac{\pi\sigma}{\cosh^{-1}\left(\frac{D}{2a}\right)}$$

(S/m)

(3.2-7)

R of 2-Wire TL

Assume current J_s (A/m) flows in very thin surface layer

$\therefore$ current in each wire: $I = 2\pi a J_s$. ($2\pi a$ being circumference)

Refer back to (2.2-9): power per unit length: $P_\sigma = 0.5 I^2 (R_s/w)$, where I is total surface current on flat conducting plate of width w .

Instead of $P_\sigma = w p_\sigma$ earlier for flat conducting plate of width w , (p_σ being power density in W/m²), now we have (for cylind. wires):

- $P_\sigma = (2\pi a) p_\sigma$ for curved circumferential surface
- Eq. (2.2-9) now translates to: $P_\sigma = 0.5 I^2 [R_s/(2\pi a)]$, i.e.,:

$$P_\sigma = 2\pi a p_\sigma = \frac{1}{2} I^2 \frac{R_s}{2\pi a} \overset{\text{write out}}{=} \left(\frac{I}{2} \right)^2 \underbrace{R}_{\substack{\text{series} \\ \text{resistance} \\ \text{per unit} \\ \text{length} \\ \text{for both} \\ \text{wires}}} \quad (\text{W/m}) \quad \textbf{(3.2-8)}$$

$$\therefore R = 2 \frac{R_s}{2\pi a} = \frac{1}{\pi a} \sqrt{\overbrace{\pi f \mu_c}^{\substack{\text{from (2.2-7c)} \\ R_s}}} \sigma_c \quad (\Omega / \text{m}) \quad \textbf{(3.2-9)}$$

Coaxial Transmission Line (C , L & G)

Radius of solid inner (center) conducting rod = a ,

Inner radius of outer cylindrical conducting pipe = b .

$$L = \frac{\mu}{2\pi} \ln \frac{b}{a} \quad (\text{H/m}) \quad (3.2-10)$$

$$C = \frac{2\pi\epsilon}{\ln\left(\frac{b}{a}\right)} \quad (\text{F/m}) \quad (3.2-11)$$

$$G = \frac{2\pi\sigma}{\ln\left(\frac{b}{a}\right)} \quad (\text{S/m}) \quad (3.2-12)$$

where the latter σ = conductivity of lossy dielectric; which could be replaced by $\omega\epsilon''$, as in (7.29) of Chapter 1, which states $\sigma = \omega\epsilon''$ (S/m)³⁴

R of Coaxial TL

Assume current J_{si} on surface of inner rod, J_{so} on surface of outer pipe $\rightarrow$ must have $I = 2\pi a J_{si} = 2\pi b J_{so}$.

Refer back to (2.2-9): power per unit length: $P_\sigma = 0.5I^2(R_s/w)$, where I is total surface current on flat conducting plate of width w .

Instead of $P_\sigma = wp_\sigma$ earlier for flat conducting plate of width w , (p_σ being power density in W/m²), now we have (for cylind. surfaces):

- $P_{\sigma i} = (2\pi a)p_{\sigma i}$ & $P_{\sigma o} = (2\pi b)p_{\sigma o}$ for curved circumferential surface
- Eq. (2.2-9) becomes: $P_{\sigma i} = 0.5I^2[R_s/(2\pi a)]$ & $P_{\sigma o} = 0.5I^2[R_s/(2\pi b)]$

$$\text{total power : } P_\sigma = P_{\sigma i} + P_{\sigma o} = \frac{I^2}{2} \frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right) \quad (\text{W/m}) \quad \textbf{(3.2-13)}$$

$$\therefore R = \frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right) = \frac{1}{2\pi} \overbrace{\sqrt{\frac{\pi f \mu_c}{\sigma_c}}}^{\text{from (2.2-7c)} \atop R_s} \left(\frac{1}{a} + \frac{1}{b} \right) \quad (\Omega/\text{m}) \quad \textbf{(3.2-14)}$$

Summary Table of Parameters for Two-Wire & Coaxial Transmission Lines

TABLE 9-2

Distributed Parameters of Two-Wire and Coaxial Transmission Lines

Parameter	Two-Wire Line	Coaxial Line	Unit
R	$\frac{R_s}{\pi a}$	$\frac{R_s}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$	Ω/m
L	$\frac{\mu}{\pi} \cosh^{-1} \left(\frac{D}{2a} \right)$	$\frac{\mu}{2\pi} \ln \frac{b}{a}$	H/m
G	$\frac{\pi\sigma}{\cosh^{-1} (D/2a)}$	$\frac{2\pi\sigma}{\ln (b/a)}$	S/m
C	$\frac{\pi\epsilon}{\cosh^{-1} (D/2a)}$	$\frac{2\pi\epsilon}{\ln (b/a)}$	F/m

Note: $R_s = \sqrt{\pi f \mu_c / \sigma_c}$; $\cosh^{-1} (D/2a) \cong \ln (D/a)$ if $(D/2a)^2 \gg 1$. Internal inductance is not included.

Part 3.3: Attenuation Constant from Power Relations

From Power Relation

- Attenuation constant of traveling wave on TL = real part of propagation constant:

$$\alpha = \operatorname{Re}(\gamma) \stackrel{\text{use}}{(3.8)} \operatorname{Re} \left[\sqrt{(R + j\omega L)(G + j\omega C)} \right] \quad (3.3-1)$$

- Can also be found from power relationship; for +z traveling, phasor voltage & current on TL [from (3.9a) & (3.9b)]:

$$\checkmark V(z) = V^+(z) + V^-(z) = V_0^+ e^{-(\alpha + j\beta)z} + V_0^- e^{+(\alpha + j\beta)z}, \quad (3.3-2a)$$

$$\checkmark I(z) = I^+(z) + I^-(z) = \underbrace{I_0^+}_{V_0^+/Z_0 \text{ from (3.15) earlier}} e^{-(\alpha + j\beta)z} + I_0^- e^{+(\alpha + j\beta)z}, \quad (3.3-2b)$$

Time-Average Power Propagated along TL at any z :

Using (3.3-2a&b) & drop + superscript:

$$\begin{aligned}
 P(z) &= \frac{1}{2} \operatorname{Re} \left[\underbrace{V(z)}_{\substack{V_0 e^{-(\alpha+j\beta)z} \\ = V_0 e^{-\alpha z} e^{-j\beta z}}} \underbrace{I^*(z)}_{\substack{\left[\frac{V_0}{Z_0} e^{-(\alpha+j\beta)z} \right]^* \\ = \frac{V_0^* e^{-\alpha z} e^{j\beta z}}{Z_0^*}}} \right] = \frac{\overbrace{|V_0|^2}^{V_0 V_0^*}}{2 |Z_0|^2} \underbrace{\operatorname{Re}(R_0 + jX_0)}_{R_0} e^{-2\alpha z} = \frac{|V_0|^2}{2 |Z_0|^2} R_0 e^{-2\alpha z} \quad (3.3-3) \\
 &= \frac{\underbrace{Z_0^*}_{R_0 - jX_0}}{\underbrace{(R_0 - jX_0)(R_0 + jX_0)}_{|Z_0|^2}} V_0^* e^{-\alpha z} e^{j\beta z} (R_0 + jX_0)
 \end{aligned}$$

Energy Conservation Law

Rate of decrease of $P(z)$ with distance along line =
time-average power loss P_L per unit length

$$\underbrace{-\frac{\partial \overbrace{P(z)}^{\text{above}}}{\partial z}}_{2\alpha P(z)} \overset{\text{equated to}}{=} \underbrace{P_L(z)}_{\substack{\text{time-average} \\ \text{power loss} \\ \text{per unit length}}} \Rightarrow \boxed{\alpha = \frac{P_L(z)}{2P(z)} \quad (\text{Np/m})} \quad (3.3-4)$$

Example: Atten Const of Lossy TL (1)

- Use Eq. (3.3-4) to find attenuation constant of lossy TL with distributed parameters R , L , G , & C .
- Specialize result in part (a) to obtain attenuation constants of low-loss line

Solution

- For lossy TL, time-average power loss per unit length:

$$P_L(z) = \frac{1}{2} \left[\underbrace{|I(z)|^2}_{\substack{\frac{|V_0|^2}{|Z_0|^2} e^{-2\alpha z} \\ \text{using} \\ (3.3-2b)}} R + \underbrace{|V(z)|^2}_{\substack{|V_0|^2 e^{-2\alpha z} \\ \text{using} \\ (3.3-2a)}} G \right] = \frac{|V_0|^2}{2|Z_0|^2} \left(R + G|Z_0|^2 \right) e^{-2\alpha z} \quad (3.3-5)$$

Substitute (3.3-3) & this (3.3-5) into (3.3-4):

$$\alpha = \frac{\overbrace{\frac{|V_0|^2}{2|Z_0|^2} \left(R + G|Z_0|^2 \right) e^{-2\alpha z}}^{P_L(z)}}{\underbrace{2 \frac{|V_0|^2}{2|Z_0|^2} R_0 e^{-2\alpha z}}_{P(z)}} = \frac{R + G|Z_0|^2}{2R_0} \quad (\text{Np/m}) \quad (3.3-6)$$

Example: Atten Const of Lossy TL (2)

(b) For low-loss line, from (3.26): $Z_0 \cong R_0 = \sqrt{L/C}$

Then, (3.3-6) becomes:

$$\alpha \cong \frac{\underbrace{\cancel{\sqrt{\frac{L}{C}}} \left(R \sqrt{\frac{C}{L}} + G \sqrt{\frac{L}{C}} \right)}_{L/C}}{2 \underbrace{R_0}_{\cancel{\sqrt{L/C}}}} = \frac{1}{2} \left(R \sqrt{\frac{C}{L}} + G \sqrt{\frac{L}{C}} \right) \quad (3.3-7)$$

which checks with (3.24a) earlier.

Part 4: Wave Characteristics on Finite Transmission Lines

Opening Remarks

- In Part 3.1, general solutions for time-harmonic 1D Helmholtz's Eqs. (3.7a) & (3.7b) for TLs:

$$\checkmark V(z) = V^+(z) + V^-(z) = V_0^+ e^{-\gamma z} + V_0^- e^{+\gamma z}, \quad (4.1a)$$

$$\checkmark I(z) = I^+(z) + I^-(z) = I_0^+ e^{-\gamma z} + I_0^- e^{+\gamma z}, \quad (4.1b)$$

- where from (3.15):
$$\frac{V_0^+}{I_0^+} = -\frac{V_0^-}{I_0^-} = Z_0 \quad (4.2)$$
- + & – superscripts: +z & –z directions of wave travel

Infinitely-Long & Matched TLs

- For waves launched towards $+z$ direction on infinitely-long line at $z = 0$,
 - only forward waves traveling in that $+z$ direction
 - vanishing 2nd terms on RHS of (4.1a) & (4.1b) with “ $-$ ” superscripts representing reflected waves
- These phenomena also true for finite lines terminated in characteristic impedance, i.e. when TLs *matched*.
- Line matched when load impedance = characteristic impedance of line

Load-Terminated Finite-Length TL

- Consider now, general case of finite TL, length ℓ with characteristic impedance Z_0 terminated in arbitrary load impedance Z_L .
- Sinusoidal voltage source: $V_g \angle 0^\circ$ with internal impedance Z_g connected to line at $z = 0$

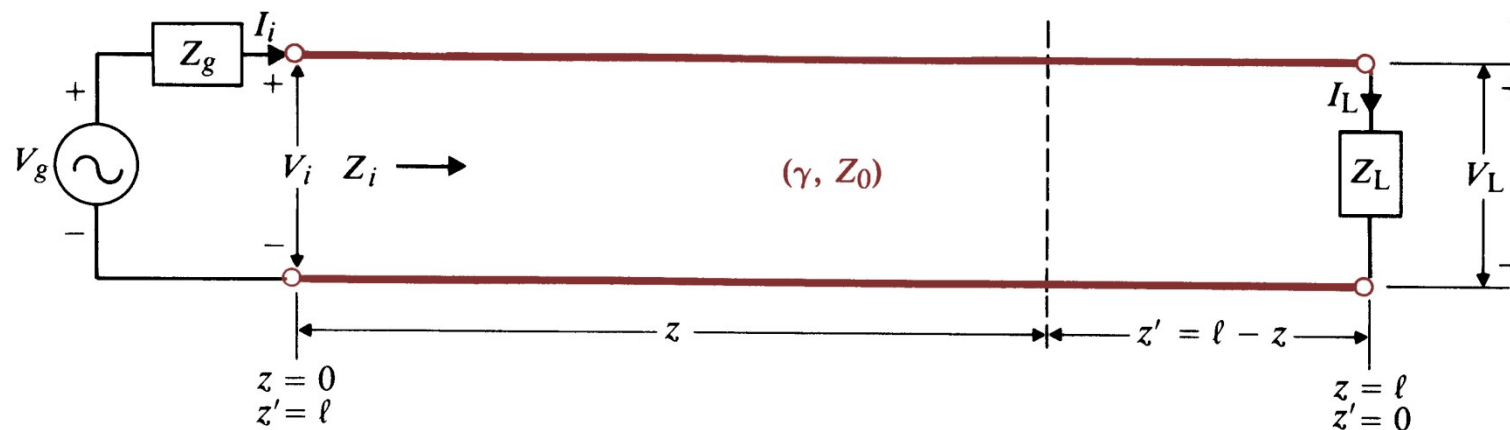


FIGURE 9-6

Finite transmission line terminated with load impedance Z_L .

Reflected Waves on Unmatched TLs

- Impedance at load end: $\left(\frac{V}{I}\right)_{z=\ell} = \frac{\overbrace{V_L}^{\text{voltage @loadend}}}{\underbrace{I_L}_{\text{current @loadend}}} = Z_L \quad (4.3)$

- This above cannot be satisfied without 2nd terms on RHS of (4.1a) & (4.1b), $\because$ if without them,

$$\frac{V(z)}{I(z)} \overset{\substack{\text{use (4.1a) \& (4.1b) \\ \text{with null 2nd terms} \\ \text{on RHS,} \\ \text{i.e. no reflected waves}}}}{=} \frac{V_0^+}{I_0^+} \overset{\text{by (4.2)}}{=} Z_0 \quad (4.4a)$$

$$\left.\frac{V(z)}{I(z)}\right|_{z=\ell} = \frac{V_L}{I_L} = Z_L = \text{load imped @ } z = \ell \quad (4.4b)$$

By (4.4a) above if no reflected waves, imped stays fixed at constant Z_0 throughout TL (all z), including at load end, $z = \ell$, BUT paradox since would be Z_L there, which is generally different from Z_0

reflected waves exist on unmatched lines

Remarks

- Given γ & Z_0 of the line & its length $\rightarrow$ 4 unknowns V_0^+ , V_0^- , I_0^+ , I_0^- in (4.1a) & (4.1b)
- These 4 unknowns not all independent $\because$ constrained by relations at $z = 0$ & $z = \ell$
- Both $V(z)$ & $I(z)$ expressible in terms of V_i & I_i at input end, or in terms of conditions at load end
- Consider latter case ...
-

Working (1)

Let $z = \ell$ in (4.1a) & (4.1b), using also (4.2):

$$V_L = V_0^+ e^{-\gamma\ell} + V_0^- e^{\gamma\ell} \quad (4.5a)$$

$$I_L = \frac{V_0^+}{Z_0} e^{-\gamma\ell} - \frac{V_0^-}{Z_0} e^{\gamma\ell} \quad (4.5b)$$

Solving these (4.5a) & (4.5b) for V_0^+ , V_0^- :

$$\text{From (4.5a): } V_0^+ = V_L e^{\gamma\ell} - V_0^- e^{2\gamma\ell} \quad (1)$$

$$\text{From (4.5b): } V_0^- = V_0^+ e^{-2\gamma\ell} - I_L Z_0 e^{-\gamma\ell} \quad (2)$$

$$\text{Sub (2) into (1): } V_0^+ = V_L e^{\gamma\ell} - (V_0^+ e^{-2\gamma\ell} - I_L Z_0 e^{-\gamma\ell}) e^{2\gamma\ell} = V_L e^{\gamma\ell} - V_0^+ + I_L Z_0 e^{\gamma\ell}$$

$$\Rightarrow 2V_0^+ = e^{\gamma\ell} (V_L + I_L Z_0) \Rightarrow V_0^+ = \frac{1}{2} (V_L + I_L Z_0) e^{\gamma\ell} \quad (4.6a)$$

$$\text{Sub (1) into (2): } V_0^- = e^{-2\gamma\ell} (V_L e^{\gamma\ell} - V_0^- e^{2\gamma\ell}) - I_L Z_0 e^{-\gamma\ell} = V_L e^{-\gamma\ell} - V_0^- - I_L Z_0 e^{-\gamma\ell}$$

$$\Rightarrow 2V_0^- = e^{-\gamma\ell} (V_L - I_L Z_0) \Rightarrow V_0^- = \frac{1}{2} (V_L - I_L Z_0) e^{-\gamma\ell} \quad (4.6b)$$

Working (2)

From (4.6a): $V_0^+ = \frac{1}{2}(\underbrace{V_L}_{\substack{I_L Z_L \\ \text{from} \\ (4.3)}} + I_L Z_0)e^{\gamma\ell} = \frac{I_L e^{\gamma\ell}}{2}(Z_L + Z_0) \quad (3)$

From (4.6b): $V_0^- = \frac{1}{2}(\underbrace{V_L}_{\substack{I_L Z_L \\ \text{from} \\ (4.3)}} - I_L Z_0)e^{-\gamma\ell} = \frac{I_L e^{-\gamma\ell}}{2}(Z_L - Z_0) \quad (4)$

Sub these (3) & (4) into (4.1a):

$$V(z) = \overbrace{\frac{I_L e^{\gamma\ell}}{2}(Z_L + Z_0)e^{-\gamma z}}^{V_0^+} + \overbrace{\frac{I_L e^{-\gamma\ell}}{2}(Z_L - Z_0)e^{\gamma z}}^{V_0^-} = \frac{I_L}{2}[(Z_L + Z_0)e^{\gamma(\ell-z)} + (Z_L - Z_0)e^{-\gamma(\ell-z)}] \quad (4.7a)$$

From (4.2): $I_0^+ = \frac{V_0^+}{Z_0} \stackrel{\text{sub}}{\underset{(3)}{=}} \frac{I_L e^{\gamma\ell}}{2Z_0}(Z_L + Z_0) \quad (5)$

& also: $I_0^- = -\frac{V_0^-}{Z_0} \stackrel{\text{sub}}{\underset{(4)}{=}} -\frac{I_L e^{-\gamma\ell}}{2Z_0}(Z_L - Z_0) \quad (6)$

Sub (5) & (6) into (4.1b):

$$I(z) = \overbrace{\frac{I_L e^{\gamma\ell}}{2Z_0}(Z_L + Z_0)e^{-\gamma z}}^{V_0^+/Z_0} - \overbrace{\frac{I_L e^{-\gamma\ell}}{2Z_0}(Z_L - Z_0)e^{\gamma z}}^{V_0^-/Z_0} = \frac{I_L}{2Z_0}[(Z_L + Z_0)e^{\gamma(\ell-z)} - (Z_L - Z_0)e^{-\gamma(\ell-z)}] \quad (4.7b)$$

New Variable: $z' =$ distance measured backward from load

Since ℓ & z appear together as $(\ell - z)$, expedient to introduce new variable: $z' = \ell - z$, being distance measured backward from load

$\therefore$ Eqs. (4.7a) & (4.7b) become:

$$V(z') = \frac{I_L}{2} \left[(Z_L + Z_0) e^{\gamma z'} + (Z_L - Z_0) e^{-\gamma z'} \right] \quad (4.8a)$$

$$I(z') = \frac{I_L}{2Z_0} \left[(Z_L + Z_0) e^{\gamma z'} - (Z_L - Z_0) e^{-\gamma z'} \right] \quad (4.8b)$$

Use of hyperbolic functions: $e^{\gamma z'} + e^{-\gamma z'} = 2 \cosh \gamma z'$ & $e^{\gamma z'} - e^{-\gamma z'} = 2 \sinh \gamma z'$
these (4.8a) & (4.8b) rewritten as:

These can be used to find voltage & current at any point along TL in terms of I_L , Z_L , γ & Z_0 .

$$V(z') = \frac{I_L}{2} \left[\underbrace{(Z_L + Z_0) e^{\gamma z'}}_{\substack{2 \cosh \gamma z' \\ Z_L(e^{\gamma z'} + e^{-\gamma z'}) + Z_0(e^{\gamma z'} - e^{-\gamma z'})}} + \underbrace{(Z_L - Z_0) e^{-\gamma z'}}_{2 \sinh \gamma z'} \right] = I_L (Z_L \cosh \gamma z' + Z_0 \sinh \gamma z') \quad (4.9a)$$

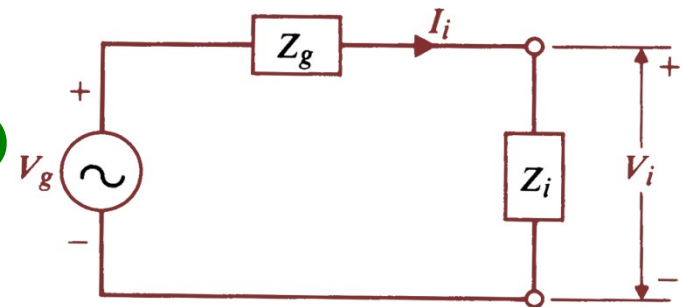
$$I(z') = \frac{I_L}{2Z_0} \left[\underbrace{(Z_L + Z_0) e^{\gamma z'}}_{2 \sinh \gamma z'} - \underbrace{(Z_L - Z_0) e^{-\gamma z'}}_{\substack{2 \cosh \gamma z' \\ Z_L(e^{\gamma z'} - e^{-\gamma z'}) + Z_0(e^{\gamma z'} + e^{-\gamma z'})}} \right] = \frac{I_L}{Z_0} (Z_L \sinh \gamma z' + Z_0 \cosh \gamma z') \quad (4.9b)$$

Input Impedance

Ratio $V(z')/I(z') =$ impedance looking towards load end of TL at distance z' from load

$$Z(z') = \frac{V(z')}{I(z')} = \frac{\overbrace{I_L (Z_L \cosh \gamma z' + Z_0 \sinh \gamma z')}^{V(z')}}{\underbrace{\frac{I_L}{Z_0} (Z_L \sinh \gamma z' + Z_0 \cosh \gamma z')}_{I(z')}} = Z_0 \frac{Z_L \cosh \gamma z' + Z_0 \sinh \gamma z'}{Z_L \sinh \gamma z' + Z_0 \cosh \gamma z'} \quad (4.10a)$$

$$\therefore Z(z') = Z_0 \frac{Z_L + Z_0 \tanh \gamma z'}{Z_0 + Z_L \tanh \gamma z'} \quad (\Omega) \quad (4.10b)$$



At source end of TL, $z' = \ell$, generator looking into line sees an *input impedance* Z_i :

$$Z_i = Z(z' = \ell; z = 0) = Z_0 \frac{Z_L + Z_0 \tanh \gamma \ell}{Z_0 + Z_L \tanh \gamma \ell} \quad (\Omega) \quad (4.11)$$

As far as generator concerned, terminated finite TL can be replaced by Z_i .

Input Voltage & Current

- Input voltage V_i & input current I_i in Fig. 9-6 earlier found from equivalent circuit in latter circuit figure by basic circuit theory:

$$V_i = \frac{Z_i}{Z_g + Z_i} V_g \quad (4.12a)$$

$$I_i = \frac{V_g}{Z_g + Z_i} \quad (4.12b)$$

- Voltage & current at any other location on line cannot be determined by latter equivalent circuit

Average Power

- Average power delivered by generator to input terminals of TL:

$$P_i^{av} = \frac{1}{2} \operatorname{Re}(V_i I_i^*)_{z=0, z'=\ell} \quad (4.13)$$

- Average power delivered to load:

$$\begin{aligned}
 P_L^{av} &= \frac{1}{2} \operatorname{Re}(V_L I_L^*)_{z=\ell, z'=0} \stackrel{\text{use(4.3)}}{=} \frac{1}{2} \operatorname{Re}\left(\frac{\overbrace{V_L V_L^*}^{|V_L|^2}}{\underbrace{Z_L^*}_{R_L - jX_L}}\right) = \frac{1}{2} \operatorname{Re}\left[\frac{|V_L|^2 (R_L + jX_L)}{\underbrace{(R_L - jX_L)(R_L + jX_L)}_{R_L^2 + X_L^2 = |Z_L|^2}}\right] \\
 &= \frac{1}{2} \left|\frac{V_L}{Z_L}\right|^2 \overbrace{\operatorname{Re}(R_L + jX_L)}^{R_L} = \frac{1}{2} \left|\frac{V_L}{Z_L}\right|^2 R_L \stackrel{\text{use(4.3)}}{=} \frac{1}{2} |I_L|^2 R_L \quad (4.14)
 \end{aligned}$$

- For lossless line, conservation of power requires: $P_i^{av} = P_L^{av}$

Match Terminated TL

- Important special case: TL terminated with its characteristic impedance, i.e. $Z_L = Z_0$.
- Input impedance Z_i in (4.11) becomes equal to Z_0 , for **all z' or z** , i.e. $Z(z') = Z_0$; for $Z_L = Z_0$.
- Voltage & current equations in (4.7a) & (4.7b):

$$V(z)|_{Z_L=Z_0} = \frac{I_L}{2} \left[\underbrace{(\overbrace{Z_L}^{Z_0} + Z_0)}_{2Z_0} e^{\gamma(\ell-z)} + \underbrace{(\overbrace{Z_L}^{Z_0} - Z_0)}_0 e^{-\gamma(\ell-z)} \right] = \underbrace{I_L Z_0 e^{\gamma(\ell-z)}}_{\underbrace{I_L Z_0 e^{\gamma\ell}}_{V_i} e^{-\gamma z}} \quad (4.15a)$$

$$I(z)|_{Z_L=Z_0} = \frac{I_L}{2Z_0} \left[\underbrace{(\overbrace{Z_L}^{Z_0} + Z_0)}_{2Z_0} e^{\gamma(\ell-z)} - \underbrace{(\overbrace{Z_L}^{Z_0} - Z_0)}_0 e^{-\gamma(\ell-z)} \right] = \underbrace{I_L e^{\gamma(\ell-z)}}_{\underbrace{I_L e^{\gamma\ell}}_{I_i} e^{-\gamma z}} \quad (4.15b)$$

Matched TL

- These (4.15a) & (4.15b) represent waves traveling in $+z$ direction without reflected waves
- $\therefore$ when finite TL terminated with its own characteristic impedance, i.e. when it is matched, voltage & current distributions on the line exactly same as through line extended to infinity.

Example (#1)

A signal generator with internal resistance $1\ \Omega$ and open-circuit voltage $v_g(t) = 0.3\cos(2\pi 10^8 t)$ (V) connected to a $50\ \Omega$ TL of length 4 m. Velocity of wave propagation along TL is 2.5×10^8 m/s. For a matched load, find (a) instantaneous expressions for voltage & current at arbitrary location on line, (b) instantaneous expressions for voltage & current at the load, and (c) average power transmitted to the load.

Solution

To find V & I at any location along line, must first obtain them at input end, i.e., $z = 0$, $z' = \ell$.

Start by first listing out given quantities:

- $V_g = 0.3\angle 0^\circ$ (V) a phasor with cosine reference {the 0.3 magnitude}
- $Z_g = R_g = 1\ (\Omega)$ {“generator with internal resistance $1\ \Omega$ ”}
- $Z_0 = R_0 = 50\ (\Omega)$ {“ $50\ \Omega$ TL”}
- $\omega = 2\pi \times 10^8$ (rad/s) $\{\because \text{“}\cos(2\pi \times 10^8 t)\text{”}\}$
- $u_p = 2.5 \times 10^8$ (m/s) {“velocity of wave”}
- $\ell = 4$ (m) {“length 4 m”}

Example (#2)

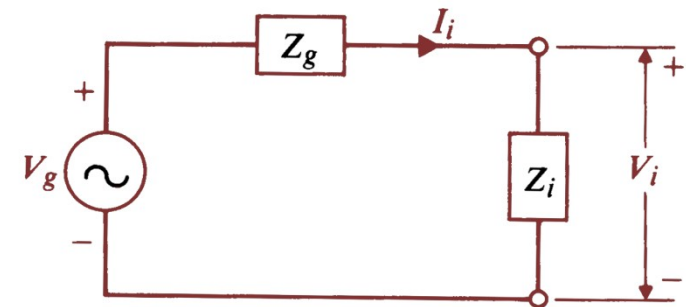
Since line terminated with matched load, $Z_i = Z_0 = 50 \Omega$, can use equivalent circuit on Slide#78 “Input Impedance” (placed below again)

From (4.12a) & (4.12b):

$$V_i = \frac{\overbrace{\frac{50}{Z_i}}^{\text{being (4.12a)}}}{\underbrace{Z_g}_{1} + \underbrace{Z_i}_{50}} \overbrace{V_g}^{0.3 \angle 0^\circ} = 0.294 \angle 0^\circ \quad (\text{V}) \quad (4.16a)$$

$$I_i = \frac{\overbrace{V_g}^{0.3 \angle 0^\circ}}{\underbrace{Z_g}_{1} + \underbrace{Z_i}_{50}} = 0.0059 \angle 0^\circ \quad (\text{A}) \quad (4.16b)$$

being (4.12b)



Example (#3)

Since only forward-traveling waves exist on matched line, use (3.3-2a) and (3.3-2b) for voltage & current, respectively, at arbitrary z-location. For given lossless line, $\alpha = 0$, with phase constant given as:

$$\beta = \frac{\omega}{u_p} = \frac{2\pi \times 10^8}{2.5 \times 10^8} = 0.8\pi \text{ (rad/m)} \quad (4.17)$$

$\therefore$ using (3.3-2a): $V(z) = \underbrace{\underbrace{\overbrace{0.294 \angle 0^\circ}^{\substack{\text{use } V_i \\ \text{in (4.16a)} \\ \text{previous} \\ \text{slide}}} \underbrace{V_0}_{\substack{\text{here} \\ \text{being} \\ \text{voltage} \\ \text{at input} \\ \text{end at} \\ z=0}}} e^{-\left(\underbrace{\overbrace{0}_{\text{here}} \alpha + j \overbrace{0.8\pi}_{\substack{\text{from} \\ \text{above}}} \beta}\right)z} = 0.294 \overbrace{\angle 0^\circ}^1 e^{-j0.8\pi z} \text{ (V)} \quad (4.18a)$

being (3.3-2a)

& using (3.3-2b): $I(z) = \frac{\underbrace{\overbrace{0.294 \angle 0^\circ}^{\substack{\text{as above}}} \underbrace{V_0}_{\substack{\text{from} \\ \text{above}}} }}{\underbrace{\underbrace{Z_0}_{50\Omega} \text{ here}}_{\text{being (3.3-2b)}}} e^{-\left(\underbrace{\overbrace{0}_{\text{here}} \alpha + j \overbrace{0.8\pi}_{\substack{\text{from} \\ \text{above}}} \beta}\right)z} = \frac{0.294 \overbrace{\angle 0^\circ}^1}{\underbrace{50}_{0.0059}} e^{-j0.8\pi z} = 0.0059 e^{-j0.8\pi z} \text{ (A)} \quad (4.18b)$

Example (#4)

These (4.18a) & (4.18b) are phasors; corresponding instantaneous forms obtained using (3.5a) & (3.5b):

$$\underbrace{v(z,t) = \operatorname{Re} \left[\begin{array}{cc} \text{from (4.18a)} & \text{given} \\ 0.294 e^{-j0.8\pi z} & 2\pi \times 10^8 \\ \overbrace{V(z)} & e^{j \overbrace{\omega} t} \end{array} \right]}_{\text{being (3.5a)}} = 0.294 \underbrace{\operatorname{Re} \left[\underbrace{e^{j\pi(2 \times 10^8 t - 0.8z)}}_{\cos[\pi(2 \times 10^8 t - 0.8z)]} \right]}_{\cos[\pi(2 \times 10^8 t - 0.8z)]} = 0.294 \cos[\pi(2 \times 10^8 t - 0.8z)] \quad \text{(V)} \quad (4.19a)$$

$$\underbrace{i(z,t) = \operatorname{Re} \left[\begin{array}{cc} \text{from (4.18b)} & \text{given} \\ 0.0059 e^{-j0.8\pi z} & 2\pi \times 10^8 \\ \overbrace{I(z)} & e^{j \overbrace{\omega} t} \end{array} \right]}_{\text{being (3.5b)}} = 0.0059 \underbrace{\operatorname{Re} \left[\underbrace{e^{j\pi(2 \times 10^8 t - 0.8z)}}_{\cos[\pi(2 \times 10^8 t - 0.8z)]} \right]}_{\cos[\pi(2 \times 10^8 t - 0.8z)]} = 0.0059 \cos[\pi(2 \times 10^8 t - 0.8z)] \quad \text{(A)} \quad (4.19b)$$

Example (#5)

Instantaneous forms of voltage & current at load ($z = \ell$) obtained simply by evaluating the latter (4.19a) & (4.19b) at $z = \ell = 4$ m, i.e.,

$$\underbrace{v(z = \ell = 4, t) = 0.294 \cos[\pi(2 \times 10^8 t - 0.8 \overset{4}{\ell})]}_{\text{of (4.19a)}} = 0.294 \cos\left[\pi(2 \times 10^8 t - 3.2)\right] \quad (4.20a)$$

$$\underbrace{i(z = \ell = 4, t) = 0.0059 \cos[\pi(2 \times 10^8 t - 0.8 \overset{4}{\ell})]}_{\text{of (4.19b)}} = 0.0059 \cos\left[\pi(2 \times 10^8 t - 3.2)\right] \quad (4.20b)$$

Fact to use: average power transmitted to load on lossless line equals that at input terminal. $\therefore$ we have:

$$\begin{aligned} (P_{av})_L &= (P_{av})_i = \frac{1}{2} \operatorname{Re} \left[\underbrace{\overbrace{V(z)}^{\text{of (4.18a)}} \overbrace{I^*(z)}^{\text{conjug of (4.18b)}}}_{\text{both phasor forms}} \right] = \frac{1}{2} \operatorname{Re} \left[\underbrace{0.294 e^{-j0.8\pi z} (0.0059 e^{-j0.8\pi z})^*}_{0.294 \times 0.0059} \right] \\ &= 8.7 \times 10^{-4} \text{ (W)} = 0.87 \text{ (mW)} \quad (4.21) \end{aligned}$$

Part 4.1: Transmission Lines as Circuit Elements

Input Impedance of Lossless TL

Most cases, TL considered lossless:

$$\gamma = j\beta, \quad Z_0 = R_0, \quad \tanh \gamma \ell = \tanh(j\beta \ell) = j \tan \beta \ell$$

then (4.11) for input impedance Z_i of lossless line with length ℓ & terminated in Z_L , becomes:

$$Z_i \Big|_{\text{lossless}} = Z(z' = \ell; z = 0) = \underbrace{R_0}_{Z_0} \frac{Z_L + j \underbrace{R_0}_{Z_0} \tan \beta \ell}{\underbrace{R_0}_{Z_0} + j Z_L \tan \beta \ell} \quad (\Omega) \quad (4.1-1)$$

Four Special Cases

Case 1: Open-Circuit Termination $Z_L \rightarrow \infty$

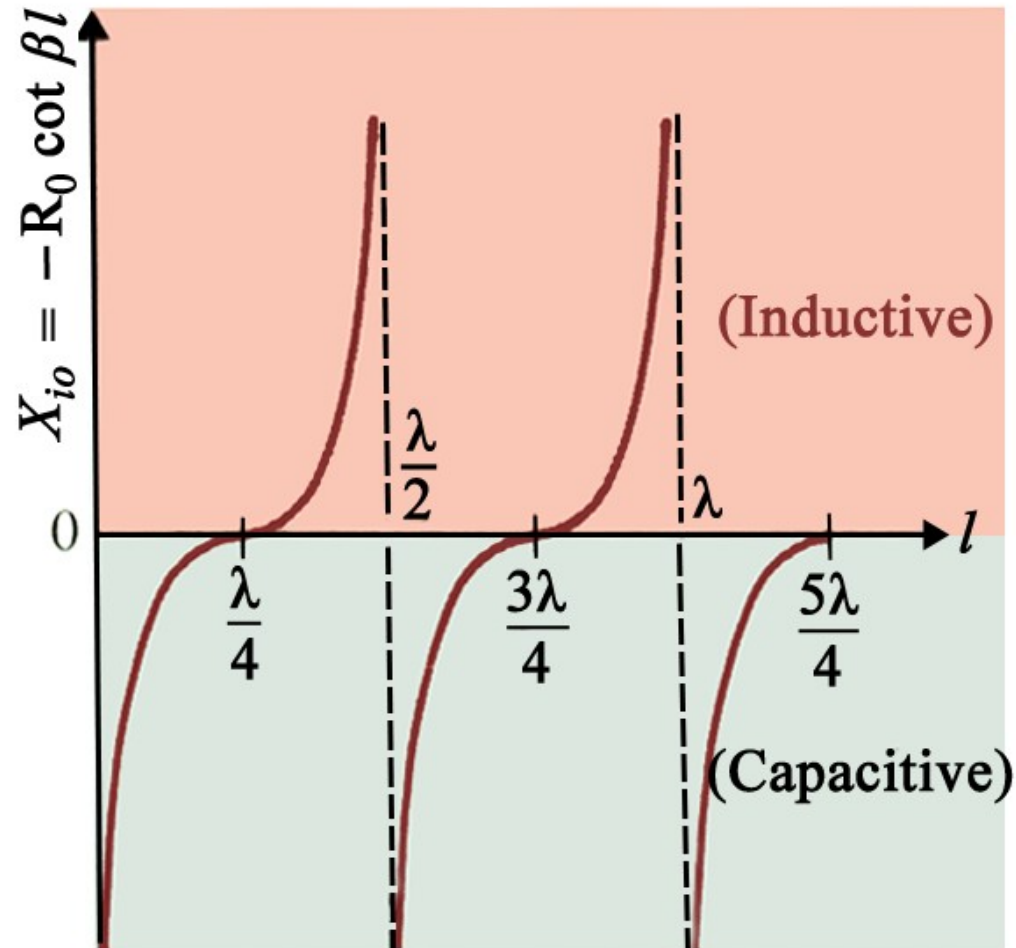
Eq. (4.1-1) yields:

$$\begin{aligned}
 Z_{io} = Z_i|_{Z_L=\infty} &= \underbrace{R_0}_{Z_0} \frac{\underbrace{\infty}_{Z_L} + j \underbrace{R_0}_{Z_0} \tan \beta \ell}{\underbrace{R_0}_{Z_0} + j \underbrace{\infty}_{Z_L} \tan \beta \ell} \stackrel{\substack{\text{divide} \\ \text{numer} \\ \& \text{denom}}}{\cong} R_0 \frac{1+0}{0+j \tan \beta \ell} \\
 &= -\frac{jR_0}{\tan \beta \ell} = -jR_0 \cot \beta \ell \stackrel{\substack{\text{assign} \\ \text{as}}}{=} j \underbrace{X_{io}}_{-R_0 \cot \beta \ell}
 \end{aligned} \tag{4.1-2}$$

- Input impedance of open-circuited lossless line purely reactive
- TL can, however, be either capacitive or inductive, $\because \cot \beta \ell$ can be either positive or negative, depending on value of $\beta \ell = 2\pi \ell / \lambda$

Graph of X_{io} versus l

Input reactance of open-circuited TL:



Indeed X_{io} ranges from $-\infty$ to $+\infty$

O. C. ($Z_L \rightarrow \infty$) with Short Length

When length of open-circuited line very short compared with wavelength, $\beta\ell \ll 1$, obtain from (4.1-2):

$$Z_{io} = jX_{io} \stackrel{\text{from (4.1-2)}}{\cong} - \underbrace{\frac{jR_0}{\tan \beta\ell}}_{\substack{\text{approx} \\ \text{as } \beta\ell \ll 1 \\ \beta\ell}} \cong -j \underbrace{\frac{R_0}{\beta}}_{\substack{\omega\sqrt{LC} \\ \text{from (3.21b)}}} \ell = -j \frac{\overbrace{R_0}^{\substack{\text{from (3.23)} \\ \sqrt{L/C}}}}{\omega\ell\sqrt{LC}} = -j \frac{1}{\omega C\ell} \quad (4.1-3)$$

being impedance of a capacitance of $C\ell$ farads

In practice, not possible to get $Z_L \rightarrow \infty$ at end of TL, esp. at high freq
 $\because$ coupling to nearby objects & radiation from open end

Case 2: Short-Circuit Termination $Z_L = 0$

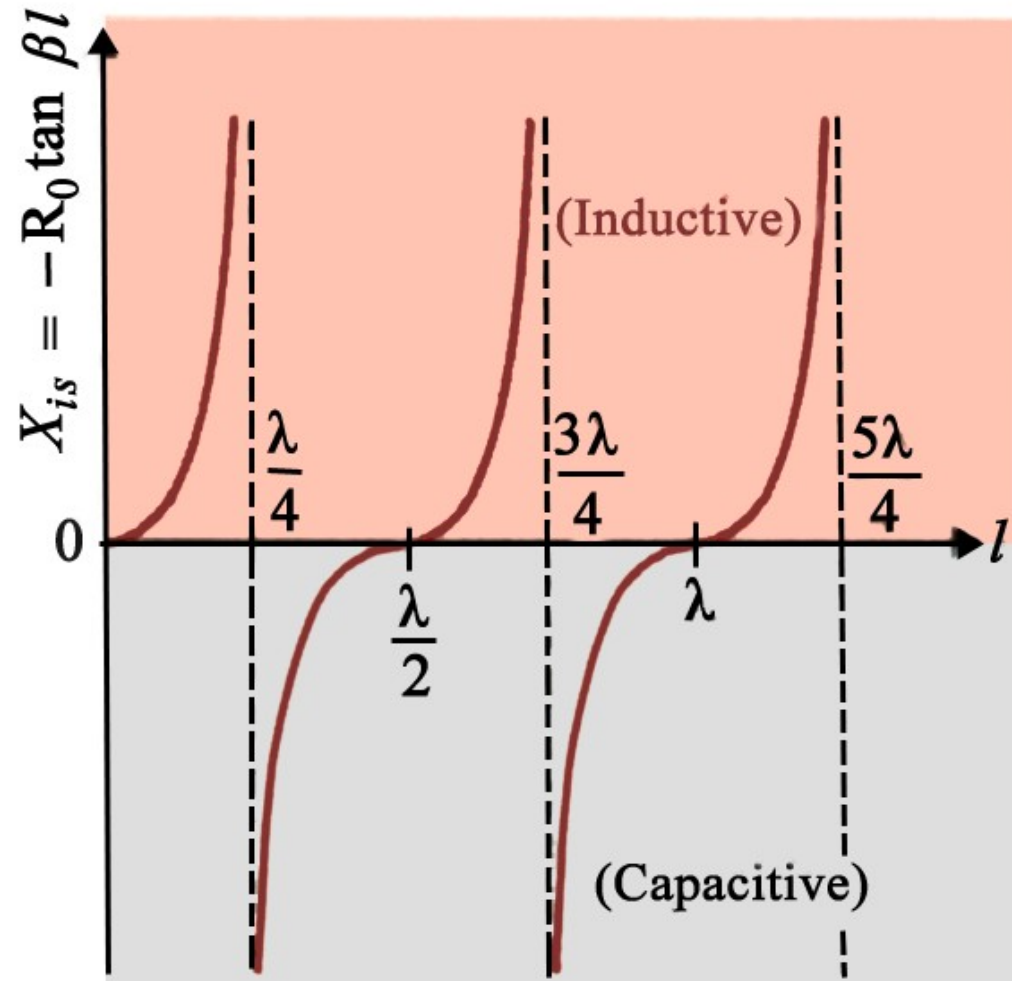
Eq. (4.1-1) yields:

$$Z_{is} = Z_i|_{Z_L=0}^{lossless} = \underbrace{R_0}_{Z_0} \frac{\underbrace{Z_L}_0 + j \underbrace{R_0}_{Z_0} \tan \beta \ell}{\underbrace{R_0}_{Z_0} + j \underbrace{Z_L}_0 \tan \beta \ell} = j \underbrace{R_0 \tan \beta \ell}_{X_{is}} \stackrel{\text{assign}}{=} j \underbrace{X_{is}}_{R_0 \tan \beta \ell} \quad (4.1-4)$$

- Input impedance of short-circuited lossless line also purely reactive (as in open-circuited)
- Same as o.c., TL can be either capacitive or inductive, $\because \tan \beta \ell$ can also be either positive or negative, depending on value of $\beta \ell = 2\pi \ell / \lambda$

Graph of X_{is} versus l

Input reactance of short-circuited TL:



Indeed X_{is} ranges from $-\infty$ to $+\infty$

S. C. ($Z_L = 0$) with Short Length

When length of short-circuited line very short compared with wavelength, $\beta\ell \ll 1$, obtain from (4.1-4):

$$Z_{is} = jX_{is} \stackrel{\text{from (4.1-4)}}{=} jR_0 \underbrace{\tan \beta\ell}_{\substack{\text{approx} \\ \text{as } \beta\ell \ll 1 \\ \beta\ell}} \cong j \underbrace{R_0}_{\substack{\sqrt{L/C} \\ \text{from (3.23)}}} \underbrace{\beta}_{\substack{\omega\sqrt{LC} \\ \text{from (3.21b)}}} \ell = j\omega L\ell \quad (4.1-5)$$

being impedance of an inductance of $L\ell$ henries

Comments

- Comparing latter 2 graphs, range where X_{io} capacitive, X_{is} inductive, & vice-versa.
- Input reactances of open-circuited & short-circuited lossless TLs same if their lengths differ by an odd multiple of $\lambda/4$.

Case 3: Quarter-Wave Section $\ell = \lambda/4; \underbrace{\beta \ell}_{\frac{2\pi}{\lambda} \frac{\lambda}{4}} = \pi/2$

When TL-length odd multiple of $\lambda/4$, $\ell = (2n-1)\lambda/4$, $n = 1, 2, 3, \dots$

$$\begin{aligned} \therefore \beta \ell &= \frac{2\pi}{\lambda} (2n-1) \frac{\lambda}{4} = (2n-1) \frac{\pi}{2} \\ \Rightarrow \tan \underbrace{\beta \ell}_{\text{above}} &= \tan \left[(2n-1) \frac{\pi}{2} \right] \rightarrow \pm\infty \end{aligned} \quad (4.1-6)$$

Placing this (4.1-6) in (4.1-1) yields:

$$\boxed{Z_i \Big|_{\substack{\text{quarterwavelength} \\ \ell=(2n-1)\frac{\lambda}{4} \\ \text{lossless}}} = \underbrace{R_0}_{Z_0} \frac{Z_L + j \underbrace{R_0}_{Z_0} \overbrace{\tan \beta \ell}^{\pm\infty}}{\underbrace{R_0}_{Z_0} + j Z_L \underbrace{\tan \beta \ell}_{\pm\infty}} = R_0 \frac{R_0}{Z_L} = \frac{R_0^2}{Z_L}} \quad (\Omega) \quad (4.1-7)$$

- $\therefore$ **quarter-wave** lossless line transforms load impedance to input terminals as its inverse multiplied by square of characteristic resistance
- Acts as impedance inverter, often called: ***quarter-wave transformer***

Case 4: Half-Wave Section $\ell = \lambda/2; \underbrace{\beta}_{\frac{2\pi}{\lambda}} \underbrace{\ell}_{\frac{\lambda}{2}} = \pi$

When TL-length integral multiple of $\lambda/2$, $\ell = n \lambda/2$, $n = 1, 2, 3, \dots$

$$\therefore \beta \ell = \frac{2\pi}{\lambda} \frac{n\lambda}{2} = n\pi \Rightarrow \tan \underbrace{\beta \ell}_{n\pi} = 0 \quad (4.1-8)$$

Placing this (4.1-8) in (4.1-1) yields:

$$\boxed{Z_i \Big|_{\substack{\text{halfwaveline} \\ \ell = n\lambda/2 \\ \text{lossless}}} = \underbrace{R_0}_{Z_0} \frac{Z_L + j \underbrace{R_0}_{Z_0} \overbrace{\tan \beta \ell}^0}{\underbrace{R_0}_{Z_0} + j Z_L \underbrace{\tan \beta \ell}_0} = Z_L \quad (\Omega)} \quad (4.1-9)$$

$\therefore$ half-wave lossless line transforms load impedance to input terminals without change

Obtain Z_0 & γ of TL by Finding Z_i under Open & Short Circuited Conditions (1)

By measuring input impedance of TL section under open & short circuit conditions, can determine characteristic impedance & propagation constant of line

Open-circuited line, $Z_L \rightarrow \infty$ in (4.11):

$$Z_{io} = Z_i|_{Z_L=\infty} = Z_0 \frac{\overbrace{Z_L}^{\infty} + Z_0 \tanh \gamma \ell}{Z_0 + \underbrace{Z_L}_{\infty} \tanh \gamma \ell} \stackrel[\substack{\text{divide} \\ \text{numerator} \\ \text{\& denominator} \\ \text{by } Z_L=\infty}]{\cong} Z_0 \frac{1 + 0}{0 + \tanh \gamma \ell} = \frac{Z_0}{\tanh \gamma \ell} = Z_0 \coth \gamma \ell \quad (4.1-10a)$$

Short-circuited line, $Z_L = 0$ in (4.11):

$$Z_{is} = Z_i|_{Z_L=0} = Z_0 \frac{\overbrace{Z_L}^0 + Z_0 \tanh \gamma \ell}{Z_0 + \underbrace{Z_L}_0 \tanh \gamma \ell} = Z_0 \tanh \gamma \ell \quad (4.1-10b)$$

Obtain Z_0 & γ of TL by Finding Z_i under Open & Short Circuited Conditions (2)

Combining these (4.1-10a) & (4.1-10b),

$$\boxed{Z_0 = \sqrt{Z_{io} Z_{is}} \quad (\Omega)} \quad (4.1-11)$$

and

$$\frac{Z_{is}}{Z_{io}} = \frac{Z_0 \tanh \gamma \ell}{Z_0 \underbrace{\coth \gamma \ell}_{\frac{1}{\tanh \gamma \ell}}} = \tanh^2 \gamma \ell \Rightarrow \tanh \gamma \ell = \sqrt{\frac{Z_{is}}{Z_{io}}} \Rightarrow \boxed{\gamma \ell = \tanh^{-1} \sqrt{\frac{Z_{is}}{Z_{io}}} \quad (\text{m}^{-1})} \quad (4.1-12)$$

These above formulas apply regardless if line lossy

Part 4.2: Lines with Resistive Termination

Opening Remarks

- When TL terminated in load impedance Z_L different from characteristic impedance Z_0 , there are both
 - incident wave (from generator) &
 - reflected wave (from load)
- Eq. (4.8a&b) give voltage & current at any distance $z' = \ell - z$ from load end
 - term $e^{\gamma z'}$: incident wave
 - term $e^{-\gamma z'}$: reflected wave

Voltage Reflection Coefficient

Repeat (4.8a):
$$V(z') = \overbrace{\frac{I_L}{2}(Z_L + Z_0)e^{\gamma z'}}^{\text{incident voltage wave}} + \overbrace{\frac{I_L}{2}(Z_L - Z_0)e^{-\gamma z'}}^{\text{reflected voltage wave}} \quad \text{(4.8a)repeated}$$

Then from this (4.8a):

$$V(z') = \frac{I_L}{2}(Z_L + Z_0)e^{\gamma z'} \left[\overbrace{1 + \frac{(Z_L - Z_0)e^{-\gamma z'}}{(Z_L + Z_0)e^{\gamma z'}}}^{\left(1 + \frac{Z_L - Z_0}{Z_L + Z_0}e^{-2\gamma z'}\right)} \right] = \frac{I_L}{2}(Z_L + Z_0)e^{\gamma z'} \left[1 + \underbrace{\left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) e^{-2\gamma z'}}_{\text{define as } \Gamma} \right] \quad \text{(4.2-1a)}$$

where

same form as (7.1.10) Chapter 2: $\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$

*for plane-wave incident
normally on plane interface
between 2 dielectric media*

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = |\Gamma| e^{j\theta_\Gamma} \quad \text{(dimensionless)} \quad \text{(4.2-1)}$$

is ratio of complex amplitudes of reflected to incident voltage waves at load ($z' = 0$) $\rightarrow$ called “**voltage reflection coefficient**” (generally complex)

Current Reflection Coefficient

Repeat (4.8b):
$$I(z') = \underbrace{\frac{I_L}{2Z_0}(Z_L + Z_0)e^{\gamma z'}}_{\text{incident current wave}} - \underbrace{\frac{I_L}{2Z_0}(Z_L - Z_0)e^{-\gamma z'}}_{\text{reflected current wave}} \quad \text{(4.8b)repeated}$$

Then from this (4.8b):
$$I(z') = \frac{I_L}{2Z_0}(Z_L + Z_0)e^{\gamma z'} \left[1 - \frac{(Z_L - Z_0)e^{-\gamma z'}}{(Z_L + Z_0)e^{\gamma z'}} \right] = \frac{I_L}{2Z_0}(Z_L + Z_0)e^{\gamma z'} \left[1 - \underbrace{\left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) e^{-2\gamma z'}}_{\text{define as } \Gamma} \right] \quad \text{(4.2-1b)}$$

$\therefore$ current reflection coefficient defined as ratio of complex amplitudes of reflected to incident current waves: I_0^-/I_0^+ , different from voltage reflection coefficient

In fact, they are negative of each other $\because I_0^-/I_0^+ = -V_0^-/V_0^+$, evident from (4.2) earlier

Shall henceforth consider only voltage reflection coefficient

Lossless Line (1)

For lossless TL, $\gamma = j\beta$; (4.2-1a) & (4.2-1b) become:

$$V(z') = \frac{I_L}{2} (Z_L + \overbrace{R_0}^{\substack{\text{loss} \\ \text{less} \\ Z_0}}) e^{\overbrace{j\beta z'}^{\substack{\text{loss} \\ \text{less} \\ \gamma}}} (1 + \underbrace{|\Gamma| e^{j\theta_\Gamma} e^{-2\overbrace{j\beta z'}^{\substack{\text{loss} \\ \text{less} \\ \gamma}}}}_{|\Gamma| e^{j(\theta_\Gamma - 2\beta z')}}) = \frac{I_L}{2} (Z_L + R_0) e^{j\beta z'} \left[1 + |\Gamma| e^{j(\theta_\Gamma - 2\beta z')} \right] \quad (4.2-2a)$$

$$I(z') = \frac{I_L}{2Z_0} (Z_L + \overbrace{R_0}^{\substack{\text{loss} \\ \text{less} \\ Z_0}}) e^{\overbrace{j\beta z'}^{\substack{\text{loss} \\ \text{less} \\ \gamma}}} (1 - \underbrace{|\Gamma| e^{j\theta_\Gamma} e^{-2\overbrace{j\beta z'}^{\substack{\text{loss} \\ \text{less} \\ \gamma}}}}_{|\Gamma| e^{j(\theta_\Gamma - 2\beta z')}}) = \frac{I_L}{2Z_0} (Z_L + R_0) e^{j\beta z'} \left[1 - |\Gamma| e^{j(\theta_\Gamma - 2\beta z')} \right] \quad (4.2-2b)$$

Lossless Line (2)

from (4.3)

Setting $\gamma = j\beta$ & $V_L = I_L Z_L$ in (4.9a) & (4.9b),

$$V(z') = \overbrace{I_L Z_L}^{V_L} \overbrace{\cosh j\beta z'}^{\cos \beta z'} + I_L \underbrace{R_0}_{\substack{Z_0 \\ \vdots \\ \text{loss} \\ \text{less}}} \overbrace{\sinh j\beta z'}^{j \sin \beta z'} = V_L \cos \beta z' + j I_L R_0 \sin \beta z' \quad (4.2-3a)$$

$$I(z') = \underbrace{\frac{I_L Z_L}{R_0}}_{\substack{Z_0 \\ \vdots \\ \text{loss} \\ \text{less}}} \overbrace{\sinh j\beta z'}^{j \sin \beta z'} + I_L \overbrace{\cosh j\beta z'}^{\cos \beta z'} = I_L \cos \beta z' + j \frac{V_L}{R_0} \sin \beta z' \quad (4.2-3b)$$

for lossless line

Voltage & Current Magnitudes

Using (4.2-3a) & (4.2-3b), voltage & current magnitudes:

$$\begin{aligned} |V(z')| &= |V_L \cos \beta z' + jI_L R_0 \sin \beta z'| = \sqrt{V_L^2 \cos^2 \beta z' + I_L^2 R_0^2 \sin^2 \beta z'} \\ &\stackrel{I_L^2 = V_L^2 / Z_L^2}{=} V_L \sqrt{\cos^2 \beta z' + \frac{R_0^2}{Z_L^2} \sin^2 \beta z'} \end{aligned} \quad (4.2-3a)$$

$$\begin{aligned} |I(z')| &= \left| I_L \cos \beta z' + j \frac{V_L}{R_0} \sin \beta z' \right| = \sqrt{I_L^2 \cos^2 \beta z' + \frac{V_L^2}{R_0^2} \sin^2 \beta z'} \\ &\stackrel{V_L^2 = I_L^2 Z_L^2}{=} I_L \sqrt{\cos^2 \beta z' + \frac{Z_L^2}{R_0^2} \sin^2 \beta z'} \end{aligned} \quad (4.2-3b)$$

where $R_0 = \sqrt{L/C}$ from (3.23).

Plots of $|V(z')|$ & $|I(z')|$ as functions of z' are standing waves with their maxima & minima occurring at fixed locations along line

Standing-Wave Ratio (SWR)

Analogously to plane-wave case in (7.1.19) of Chapter 2, repeated here:

$$S = \frac{|E|_{\max}}{|E|_{\min}} = \frac{1+|\Gamma|}{1-|\Gamma|} \quad (\text{dimensionless}) \quad (7.1.19)\text{Chapter 2}$$

define ratio of max to min voltages along finite, terminated TL as:
standing-wave ratio (SWR), S:

$$S = \frac{|V_{\max}|}{|V_{\min}|} = \frac{1+|\Gamma|}{1-|\Gamma|} \quad (\text{dimensionless}) \quad (4.2-4)$$

Same way as (7.1.20) of Chapter 2, inverse of this (4.2-4) is:

$$|\Gamma| = \frac{S-1}{S+1} \quad (\text{dimensionless}) \quad (4.2-5)$$

3 Scenarios of S & Γ

- Matched Load: $Z_L = Z_0$.
 - ✓ From (4.2-1): $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = 0$
 - ✓ From (4.2-4): $S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = 1$
- Short-circuited termination: $Z_L = 0$
 - ✓ From (4.2-1): $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = -1$
 - ✓ From (4.2-4): $S = \frac{1 + |\Gamma|}{1 - |\Gamma|} \rightarrow \infty$
- Open-circuited termination: $Z_L \rightarrow \infty$
 - ✓ From (4.2-1): $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = +1$
 - ✓ From (4.2-4): $S = \frac{1 + |\Gamma|}{1 - |\Gamma|} \rightarrow \infty$

Notes about SWR

- $\because$ wide range of S , customary to express it on logarithmic scale: $S_{\text{dB}} = 20\log_{10}S$ (dB)
- SWR S defined in terms of $|I_{\text{max}}|/|I_{\text{min}}|$ results in same expression as that defined in terms of $|V_{\text{max}}|/|V_{\text{min}}|$ in (4.2-4)
- High SWR on TL undesirable $\because$ large power loss

Simultaneous Max & Min of Voltage and Current Amplitudes

- From (4.2-2a) & (4.2-2b), seen that

✓ $|V_{\max}|$ & $|I_{\min}|$ occur together when

$$\theta_{\Gamma} - 2\beta z'_M = -2n\pi, \quad n = 0, 1, 2, \dots \quad (4.2-6a)$$

✓ $|V_{\min}|$ & $|I_{\max}|$ occur together when

$$\theta_{\Gamma} - 2\beta z'_m = -(2n + 1)\pi, \quad n = 0, 1, 2, \dots \quad (4.2-6b)$$

- Both equations repeated:

$$V(z') = \frac{I_L}{2} (Z_L + R_0) e^{j\beta z'} \left[1 + |\Gamma| e^{j(\theta_{\Gamma} - 2\beta z')} \right] \quad (4.2-2a)\text{repeated}$$

$$I(z') = \frac{I_L}{2Z_0} (Z_L + R_0) e^{j\beta z'} \left[1 - |\Gamma| e^{j(\theta_{\Gamma} - 2\beta z')} \right] \quad (4.2-2b)\text{repeated}$$

Resistive Terminations on Lossless TLs

- $Z_L = R_L$, $Z_0 = R_0$, then (4.2-1) gives:

$$\Gamma = \frac{\overbrace{R_L}^{Z_L} - \overbrace{R_0}^{Z_0}}{\overbrace{R_L}^{Z_L} + \overbrace{R_0}^{Z_0}} \rightarrow \text{purely real (resistive load)} \quad (4.2-7)$$

- Two cases:
 - ✓ $R_L > R_0$: details to follow ...
 - ✓ $R_L < R_0$: details to follow ...

Case 1: $R_L > R_0$.

- Γ positive real, $\theta_\Gamma = 0$, (4.2-8)
- $\therefore$ at termination, $z' = 0$; $z'_M = 0$ in (4.2-6a) condition (4.2-6a) satisfied for $n = 0 \rightarrow$ **voltage max** (with current min) occurs at terminating resistance
- Other **maxima of voltage standing wave** (**minima of current standing wave**) located at: {use (4.2-6a)}

$$\underbrace{\overbrace{\theta_\Gamma}^{0 \text{ here}} - 2\beta z'}_{\text{from (4.2-6a)}} = -2n\pi, \Rightarrow \underbrace{2\beta z'}_{\substack{2\pi/\lambda \\ \text{distance from load}}} = 2n\pi \Rightarrow z' = n\lambda/2 \quad n = 1, 2, \dots \quad (4.2-8a)$$

- **Minima of VSW** (**maxima of CSW**) at $(4.2-8b)$

$$\underbrace{\overbrace{\theta_\Gamma}^{0 \text{ here}} - 2\beta z'}_{\text{from (4.2-6b)}} = -(2n+1)\pi, \Rightarrow \underbrace{2\beta z'}_{\substack{2\pi/\lambda \\ 4\pi z'/\lambda \\ \beta z' = (2n+1)\pi/2}} = (2n+1)\pi \Rightarrow \frac{4z'}{\lambda} = 2n+1 \Rightarrow z' = \underbrace{\frac{(2n+1)\lambda}{4}}_{\substack{\text{distance from load}}} ; \quad n = 0, 1, 2, \dots$$

Case 2: $R_L < R_0$.

- Γ negative real, $\theta_\Gamma = -\pi$. $z'_m = 0$ in (4.2-6b) (4.2-9)
- $\therefore$ at termination, $z' = 0$; condition (4.2-6b) satisfied for $n = 0 \rightarrow$ **voltage min** (with current max) occurs at terminating resistance
- Other minima of voltage standing wave (**maxima of current standing wave**) located at:

$$\underbrace{\overbrace{\theta_\Gamma}^{-\pi} - 2\beta z'}_{\text{from (4.2-6b)}} = -(2n+1)\pi, \Rightarrow \underbrace{\pi + 2 \overbrace{\beta}^{2\pi/\lambda} z'}_{\pi \left(1 + \frac{4}{\lambda} z'\right)} = (2n+1)\pi \Rightarrow \frac{4z'}{\lambda} = 2n \Rightarrow z' = \underbrace{\frac{2n\lambda}{4} = \frac{n\lambda}{2}}_{\text{distance from load}}; \quad n = 1, 2, \dots \quad \text{(4.2-9a)}$$

- **Maxima of VSW** (**minima of CSW**) at

$$\underbrace{\overbrace{\theta_\Gamma}^{-\pi} - 2\beta z'}_{\text{from (4.2-6a)}} = -2n\pi, \Rightarrow \underbrace{\pi + 2 \overbrace{\beta}^{2\pi/\lambda} z'}_{\pi \left(1 + \frac{4}{\lambda} z'\right)} = 2n\pi \Rightarrow 1 + \frac{4}{\lambda} z' = 2n \Rightarrow z' = \underbrace{(2n-1)\lambda/4}_{\text{distance from load}} \quad n = 1, 2, \dots \quad \text{(4.2-9b)}$$

Voltage & Current Standing Waves on Resistance-Terminated Lossless Lines

- $\therefore$ roles of voltage & current standing waves interchanged between $R_L > R_0$ & $R_L < R_0$.
- Below shows some typical standing waves for lossless line with resistive termination:

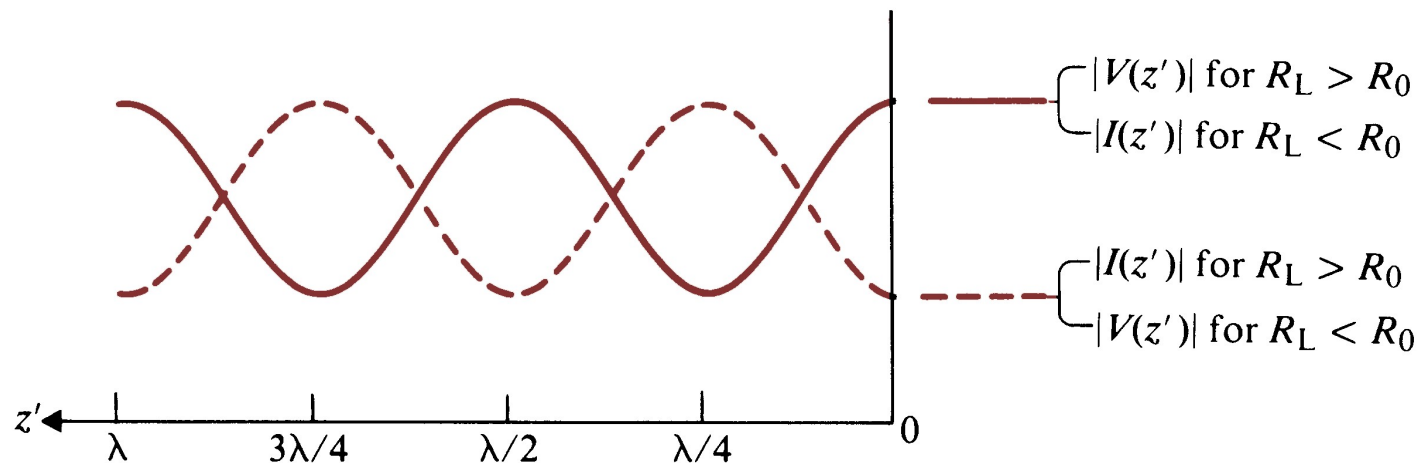


FIGURE 9-10

Voltage and current standing waves on resistance-terminated lossless lines.

Standing Waves on Short & Open Circuited Terminations

Open-Circuited Termination

- By setting $Z_L \rightarrow \infty$ in (4.2-3a) & (4.2-3b),

$$|V(z')| \stackrel{Z_L \rightarrow \infty}{\underset{\text{from (4.2-3a)}}{=}} V_L \sqrt{\cos^2 \beta z' + \frac{R_0^2}{Z_L^2} \sin^2 \beta z'} \cong V_L \sqrt{\overbrace{\cos^2 \beta z'}^{\geq 0}} = V_L |\cos \beta z'| \quad (4.2-10a)$$

cosine func $\rightarrow$ max at $z'=0$ (termination); same as $|V(z')|$ of Case 1: $R_L > R_0$.

$$|I(z')| \stackrel{Z_L \rightarrow \infty}{\underset{\text{from (4.2-3b)}}{=}} I_L \sqrt{\cos^2 \beta z' + \frac{Z_L^2}{R_0^2} \sin^2 \beta z'} \cong \frac{V_L}{R_0} |\sin \beta z'| \quad (4.2-10b)$$

simple sinusoids

sine func $\rightarrow$ min 0 at $z'=0$ (termination); same as $|I(z')|$ of Case 1: $R_L > R_0$.

- Standing waves on open-circuited line similar to those on resistance-terminated line with $R_L > R_0$ (Case 1 earlier), except that curves of $|V(z')|$ & $|I(z')|$ now magnitudes of sinusoidal functions of distance z' from load
- Since sinusoid, all minima go to 0*

Short-Circuited Termination

- By setting $Z_L = 0$ in (4.2-3a) & (4.2-3b),

$$|V(z')| \stackrel{Z_L=0}{\underset{\text{from (4.2-3a)}}{=}} V_L \sqrt{\cos^2 \beta z' + \underbrace{\frac{R_0^2}{Z_L^2} \sin^2 \beta z'}_{\infty}} \cong V_L \frac{R_0}{Z_L} \sqrt{\overbrace{\sin^2 \beta z'}^{\geq 0}} = I_L R_0 |\sin \beta z'| \quad (4.2-11a)$$

sine func $\rightarrow$ min 0 at $z'=0$ (termination); same as $|V(z')|$ of Case 2: $R_L < R_0$.

$$|I(z')| \stackrel{Z_L=0}{\underset{\text{from (4.2-3b)}}{=}} I_L \sqrt{\cos^2 \beta z' + \underbrace{\frac{Z_L^2}{R_0^2} \sin^2 \beta z'}_0} = I_L \sqrt{\overbrace{\cos^2 \beta z'}^{\geq 0}} = I_L |\cos \beta z'| \quad (4.2-11b)$$

simple sinusoids

cosine func $\rightarrow$ max at $z'=0$ (termination); same as $|I(z')|$ of Case 2: $R_L < R_0$.

- Standing waves on short-circuited line similar to those on resistance-terminated line with $R_L < R_0$ (Case 2 earlier), except that curves of $|V(z')|$ & $|I(z')|$ now magnitudes of sinusoidal functions of distance z' from load

Voltage & Current Standing Waves on Open & Short Circuited Lossless Lines

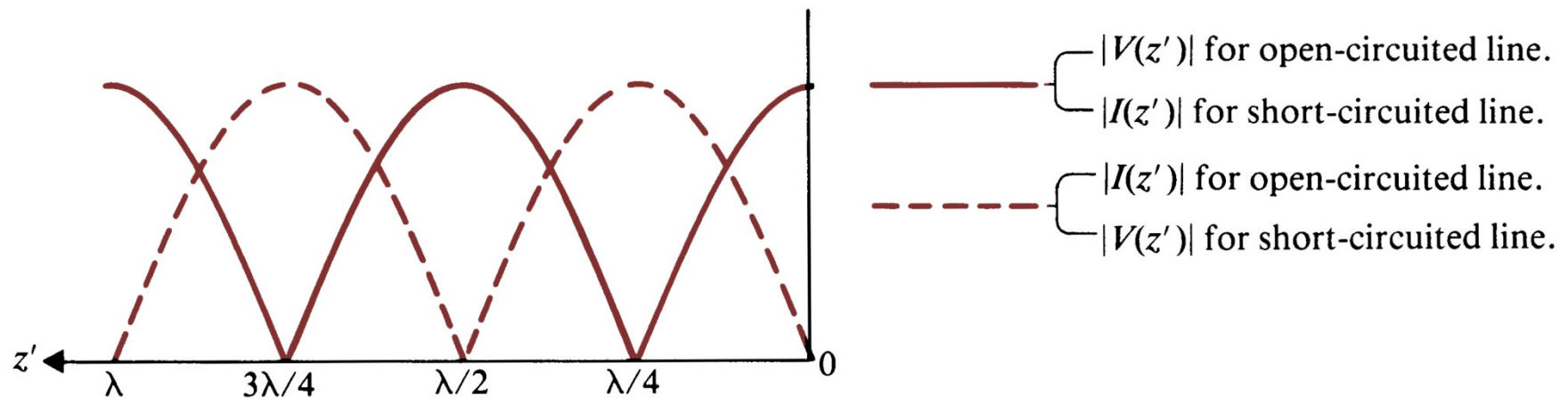


FIGURE 9-11

Voltage and current standing waves on open- and short-circuited lossless lines.

Example: SWR, Z_i , R_L of lossless TL (1)

Standing-wave ratio S on TL easily measured. (a) Show how terminating resistance on lossless line of known characteristic resistance R_0 can be determined by measuring S . (b) Find impedance of TL looking toward load at quarter-wavelength distance from it

Solution

(a) Since termination purely resistive, $Z_L = R_L$, can determine whether $R_L > R_0$ {Case 1 earlier} or $R_L < R_0$ {Case 2 earlier} (if voltage **maxima** or **minima** at $z' = 0, \lambda/2, \lambda, \dots, n\lambda/2$ etc, respectively). This easily ascertained by measurements.

First, if $R_L > R_0$ (Case 1), $\theta_\Gamma = 0 \rightarrow$ see (4.2-8)

- Both $|V_{\max}|$ & $|I_{\min}|$ occur at $z' = 0 \rightarrow$ from (4.2-8a) with $n = 0$ **Fact (1a)**
- $|V_{\min}|$ & $|I_{\max}|$ occur at $\beta z' = \pi/2 \rightarrow$ from (4.2-8b) with $n = 0$ **Fact (1b)**

Example: SWR, Z_i , R_L of lossless TL (2)

From (4.2-3a) & (4.2-3b), repeated:

$$V(z') = V_L \cos \beta z' + jI_L R_0 \sin \beta z' \quad \text{(4.2-3a)repeated}$$

$$I(z') = I_L \cos \beta z' + j \frac{V_L}{R_0} \sin \beta z' \quad \text{(4.2-3b)repeated}$$

For (4.2-3a), “Fact (1a)” previous slide yields: $|V_{\max}| = V_L$ [$\because \cos 0 = 1$ & $\sin 0 = 0$] (1a)

& “Fact (1b)” previous slide yields: $|V_{\min}| = I_L R_0 = V_L R_0 / R_L$,
[$\because \cos(\pi/2) = 0$ & $\sin(\pi/2) = 1$] (1b)

For (4.2-3b), “Fact (1a)” previous slide yields: $|I_{\min}| = I_L$ [$\because \cos 0 = 1$ & $\sin 0 = 0$] (2a)

& “Fact (1b)” previous slide yields: $|I_{\max}| = V_L / R_0 = I_L R_L / R_0$,
[$\because \cos(\pi/2) = 0$ & $\sin(\pi/2) = 1$] (2b)

both above by (4.2-4)

$$(1a)/(1b): |V_{\max}|/|V_{\min}| = R_L/R_0 = S \rightarrow R_L = S R_0. \quad \text{(4.2-12)}$$

for $R_L > R_0$ (Case 1)

Example: SWR, Z_i , R_L of lossless TL (3)

Second, if $R_L < R_0$ (Case 2), $\theta_\Gamma = -\pi \rightarrow$ see (4.2-9)

- Both $|V_{\max}|$ & $|I_{\min}|$ occur at $\beta z' = \pi/2 \rightarrow$ from (4.2-9b) with $n = 1$ **Fact (2a)**
- $|V_{\min}|$ & $|I_{\max}|$ occur at $\beta z' = 0 \rightarrow$ from (4.2-9a) with $n = 0$ **Fact (2b)**

Again from (4.2-3a) & (4.2-3b), repeated for convenience:

$$V(z') = V_L \cos \beta z' + jI_L R_0 \sin \beta z' \quad \textbf{(4.2-3a)repeated}$$

$$I(z') = I_L \cos \beta z' + j \frac{V_L}{R_0} \sin \beta z' \quad \textbf{(4.2-3b)repeated}$$

Example: SWR, Z_i , R_L of lossless TL (4)

For (4.2-3a), “Fact (2a)” previous slide yields: $|V_{\max}| = I_L R_0 = V_L R_0 / R_L$, (3a)
 & “Fact (2b)” previous slide yields: $|V_{\min}| = V_L$. (3b)
 $[\because \cos(\pi/2) = 0 \text{ \& } \sin(\pi/2) = 1]$
 $[\because \cos 0 = 1 \text{ \& } \sin 0 = 0]$

For (4.2-3b), “Fact (2a)” previous slide yields: $|I_{\min}| = V_L / R_0 = I_L R_L / R_0$, (4a)
 & “Fact (2b)” previous slide yields: $|I_{\max}| = I_L$. (4b)
 $[\because \cos(\pi/2) = 0 \text{ \& } \sin(\pi/2) = 1]$
 $[\because \cos 0 = 1 \text{ \& } \sin 0 = 0]$

both above by (4.2-4)

$$(3a)/(3b): |V_{\max}|/|V_{\min}| = R_0/R_L = S \rightarrow R_L = R_0/S. \quad \textcolor{green}{(4.2-13)}$$

for $R_L < R_0$ (Case 2)

Example: SWR, Z_i , R_L of lossless TL (5)

- Wavelength λ determined from twice distance between two neighboring voltage (or current) maxima or minima
- At $z' = \lambda/4$, $\beta z' = \pi/2$, $\cos(\beta z') = 0$ & $\sin(\beta z') = 1$.
- (4.2-3a) & (4.2-3b) become:

$$V(z' = \lambda/4) = V_L \overbrace{\cos \beta z'}^0 + j I_L R_0 \overbrace{\sin \beta z'}^1 \quad (4.2-14a)$$

$$I(z' = \lambda/4) = I_L \overbrace{\cos \beta z'}^0 + j \frac{V_L}{R_0} \overbrace{\sin \beta z'}^1 \quad (4.2-14b)$$

Example: SWR, Z_i , R_L of lossless TL (6)

$\therefore$ input impedance looking towards load at distance of $\lambda/4$ from it simply the V/I ratio using latter two equations on previous slide:

$$Z_i(z' = \lambda/4) = \frac{\overbrace{V(z' = \lambda/4)}^{\substack{\text{from (4.2-14a)} \\ jI_L R_0}}}{\underbrace{I(z' = \lambda/4)}_{\substack{j\frac{V_L}{R_0} \\ \text{from (4.2-14b)}}}} = \frac{R_0^2}{\underbrace{(V_L/I_L)}_{Z_L}} = \frac{R_0^2}{Z_L} \quad (4.2-15)$$

which is as expected from “quarter wave impedance transformer” learned earlier in (4.1-7)

Part 4.3: Lines with Arbitrary Termination

Opening Remarks

- Preceding subsection, learned about voltage & current standing waves on resistively-terminated lossless TL
- What happens if terminating impedance not pure resistance?
- Intuitively expect neither max nor min of voltage & current occur at termination; both shifted away from termination
- Here, shall show that info on direction & amount of this shift can determine terminating impedance

Preliminaries (1)

- Let terminating (or load) impedance, $Z_L = R_L + jX_L$.
- Assume voltage standing wave on line as below
- Neither min nor max at load ($z' = 0$)

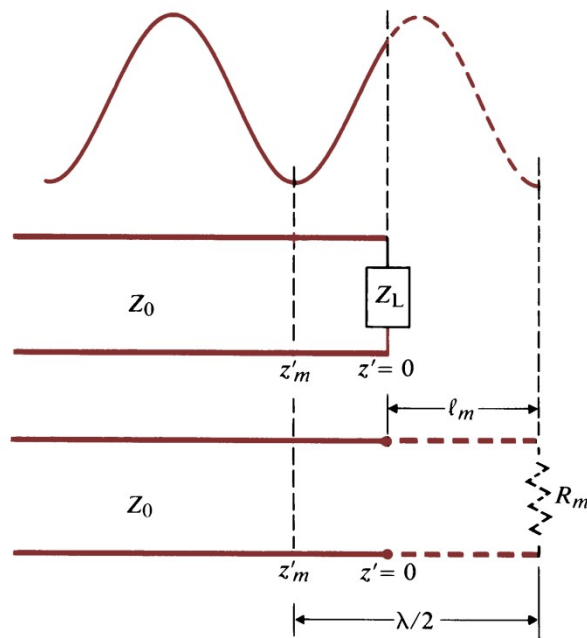
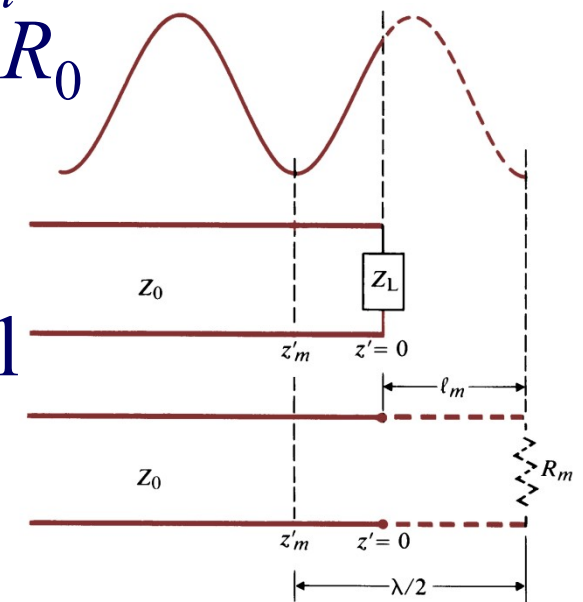


FIGURE 9-12

Voltage standing wave on a line terminated by an arbitrary impedance, and equivalent line section with pure resistive load.

Preliminaries (2)

- If let standing wave continue by extra distance l_m beyond termination, will reach voltage minimum
- Voltage **min** is where it should be if original terminating impedance Z_L replaced by line section of length l_m terminated by pure resistance $R_m < R_0$ (see figure) – “Case 2: $R_L < R_0$ ” earlier [slide containing (4.2-9)]
- Voltage distribution to left of actual Z_L termination ($z' > 0$) not changed by this replacement



Solve for R_m & ℓ_m by Simul. Eq.

- From (4.1-1): any complex impedance can be obtained as input impedance of section of lossless line terminated in resistive load.
- Using R_m for Z_L & ℓ for ℓ_m in that (4.1-1):

$$Z_i \Big|_{\text{lossless}} = \overbrace{Z(z'' = \ell_m)}^{Z_L = Z(z'=0) \text{ of (4.3-2)}} = R_i + jX_i = R_0 \frac{R_m + jR_0 \tan \beta \ell_m}{R_0 + jR_m \tan \beta \ell_m} \quad (4.3-1)$$

double-primed z measured from fictitious load R_m at end of extended section of length ℓ_m beyond $z' = 0$

- Real & imaginary parts of this (4.3-1) form two equations, from which the two unknowns R_m & ℓ_m can be solved ... *done next slide* ...

Equate real & imag parts: R_i & X_i in terms of R_m & ℓ_m .

Working on (4.3-1)

$$R_i + jX_i = R_0 \frac{R_m + jR_0 \tan \beta \ell_m}{R_0 + jR_m \tan \beta \ell_m} \overbrace{\left(\frac{R_0 - jR_m \tan \beta \ell_m}{R_0 - jR_m \tan \beta \ell_m} \right)}^1$$

$$= R_0 \frac{R_m R_0 (1 + \tan \beta \ell_m) + j \tan \beta \ell_m (R_0^2 - R_m^2)}{(R_0^2 + R_m^2 \tan^2 \beta \ell_m)}$$

$$\therefore R_i = \frac{R_m R_0^2 (1 + \tan \beta \ell_m)}{(R_0^2 + R_m^2 \tan^2 \beta \ell_m)} \quad (4.3-1a)$$

$$\& X_i = \frac{R_0 \tan \beta \ell_m (R_0^2 - R_m^2)}{(R_0^2 + R_m^2 \tan^2 \beta \ell_m)} \quad (4.3-1b)$$

Solve Simult. Eq. for R_m & ℓ_m .

From (4.3-1b)

$$X_i = \frac{R_0 \tan \beta \ell_m (R_0^2 - R_m^2)}{(R_0^2 + R_m^2 \tan^2 \beta \ell_m)}$$

$$\Rightarrow X_i R_0^2 + X_i R_m^2 \tan^2 \beta \ell_m = R_0^3 \tan \beta \ell_m - R_m^2 R_0 \tan \beta \ell_m$$

$$\Rightarrow (X_i + R_0) R_m^2 \tan^2 \beta \ell_m - R_0^3 \tan \beta \ell_m + X_i R_0^2 = 0$$

$$\Rightarrow R_m = \sqrt{\frac{R_0^2 (R_0 \tan \beta \ell_m - X_i)}{(X_i + R_0) \tan^2 \beta \ell_m}} = \frac{R_0}{\tan \beta \ell_m} \sqrt{\frac{R_0 \tan \beta \ell_m - X_i}{X_i + R_0}}$$

(4.3-1bi)

$\therefore R_m$ obtained once ℓ_m known

Then, can sub this R_m in terms of only ℓ_m into (4.3-1a) to solve for ℓ_m .

Experim. Get Load Imped. by Measuring SWR & 1st Volt. Min Nearest to Load (#1)

- Load impedance Z_L can be determined experimentally by measuring standing-wave ratio S & distance z'_m in Fig. 9-12, being distance from termination to nearest minimum

- Note that $z'_m + \ell_m = \lambda/2$

- Procedure:

- Find $|\Gamma|$ from S : Use $|\Gamma| = (S - 1)/(S + 1)$ from (4.2-5)
- Find θ_Γ from z'_m . Use $\theta_\Gamma = 2\beta z'_m - \pi$ from (4.2-6b) with $n = 0$
- Find Z_L , being ratio of (4.2-2a) & (4.2-2b), at $z' = 0$ (next slide ...)

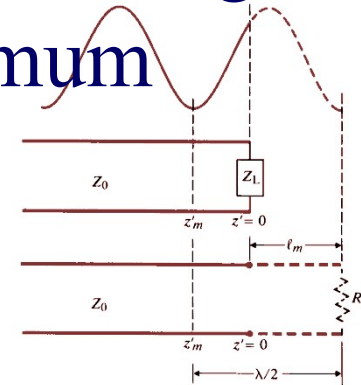


FIGURE 9-12
Voltage standing wave on a line terminated by an arbitrary impedance, and equivalent line section with pure resistive load.

Experim. Get Load Imped. by Measuring SWR & 1st Volt. Min Nearest to Load (#2)

From (4.2-2a) with $z' = 0$:

$$V(z' = 0) = \frac{I_L}{2} (Z_L + R_0) \underbrace{e^{j\beta \overset{0}{z'}}}_1 \left[1 + |\Gamma| e^{j(\theta_\Gamma - 2\beta \overset{0}{z'})} \right] = \frac{I_L}{2} (Z_L + R_0) (1 + |\Gamma| e^{j\theta_\Gamma})$$

From (4.2-2b) with $z' = 0$:

$$I(z' = 0) = \frac{I_L}{2Z_0} (Z_L + R_0) \underbrace{e^{j\beta \overset{0}{z'}}}_1 \left[1 - |\Gamma| e^{j(\theta_\Gamma - 2\beta \overset{0}{z'})} \right] = \frac{I_L}{2Z_0} (Z_L + R_0) (1 - |\Gamma| e^{j\theta_\Gamma})$$

Dividing former by latter, obtain: $Z_L =$ impedance at load-end ($z' = 0$),

$$Z(z' = 0) = Z_L = \frac{V(z' = 0)}{I(z' = 0)} = \frac{\frac{I_L}{2} (Z_L + R_0) (1 + |\Gamma| e^{j\theta_\Gamma})}{\frac{I_L}{2Z_0} (Z_L + R_0) (1 - |\Gamma| e^{j\theta_\Gamma})} = \overset{Z_0}{\cancel{R_0}} \frac{1 + |\Gamma| e^{j\theta_\Gamma}}{1 - |\Gamma| e^{j\theta_\Gamma}} \quad (4.3-2)$$

in which the $|\Gamma|$ & θ_Γ found as described on previous slide

Closing Notes

- Value of R_m which, if terminated on line of length ℓ_m , will yield input impedance Z_L , at end of extension found from (4.3-1) concerned with voltage min here (Fig. 9-12):
- Since $R_m < R_0$, $R_m = R_0/S \rightarrow$ from (4.2-13) see "Case 2: $R_L < R_0$ " on slide containing (4.2-9)
- Procedure leading to (4.3-2) used to determine Z_L from measurement of S & z'_m , distance from termination to 1st voltage minimum.
- Distance from termination to voltage max, z'_M , could be used instead of z'_m .
- In this latter case, (4.2.6a) used to find θ_Γ in Step 2 earlier (2 slides ago) $\theta_\Gamma - 2\beta z'_M = -2n\pi$, $n = 0, 1, 2, \dots$ (4.2.6a)repeated

Example (#1)

The SWR on lossless $50\ \Omega$ TL terminated in unknown load impedance is 3.0. Distance between successive voltage minima is 20 cm, and the first minimum located at 5 cm from load. Find (a) reflection coefficient Γ , (b) load impedance Z_L , (c) equivalent length & terminating resistance of a line such that the input impedance equals Z_L .

Solution

Given: (1) $S = 3$, (2) $z'_m|_{n=0} = 0.05\text{ m}$.

(3) Distance between successive voltage minima is $\lambda/2$.

$$\therefore \lambda = 2 \times 0.2 = 0.4\text{ m} \quad \Rightarrow \quad \beta = 2\pi/\lambda = 2\pi/0.4 = 5\pi\text{ (rad/m)}$$

Step 1: Use (4.2-5) to find $|\Gamma|$ from SWR $S = 3$:

$$|\Gamma| = \frac{S-1}{S+1} = \frac{3-1}{3+1} = 0.5$$

Step 2: Use (4.2-6b) with $n = 0$:

$$\theta_\Gamma = 2\beta z'_m - \pi = 2(5\pi)(0.05) - \pi = -0.5\pi\text{ (rad)}$$

$\therefore$ combine outcomes of Steps 1 & 2: $\Gamma = 0.5e^{-j\pi/2} = -0.5j$.
Ans (a)

Example (#2)

(b) To get Z_L : use (4.3-2)

$$Z(z' = 0) = Z_L = \underbrace{R_0}_{50} \frac{1 + \overbrace{|\Gamma|}^{0.5} e^{j\overbrace{\theta_\Gamma}^{\frac{\pi}{2}}}}{\underbrace{1 - |\Gamma| e^{j\theta_\Gamma}}_{-j/2}} = 50 \frac{1 - 0.5j}{1 + 0.5j} \overbrace{\left(\frac{1 - 0.5j}{1 - 0.5j} \right)}^1 = 50 \frac{1 - 0.25 - j}{1 + 0.25}$$

$$= 50 \left(\frac{0.75}{1.25} - j \frac{1}{1.25} \right) = 50(0.6 - j0.8) = 30 - j40 \quad \text{Ans (b)}$$

(c) To get ℓ_m & R_m :

we know $z'_m + \ell_m = \lambda/2 \rightarrow$ see 4 slides ago

& $R_m = R_0/S \rightarrow$ see 2 slides ago

$$\therefore \ell_m = \frac{\lambda}{2} - z'_m = 0.2 - 0.05 = 0.15 \text{ (m)}$$

$$R_m = \frac{50}{3} = 16.7 \text{ } \Omega$$

End of Chapter

*Read up the rest in textbook not
covered in lectures on your own*